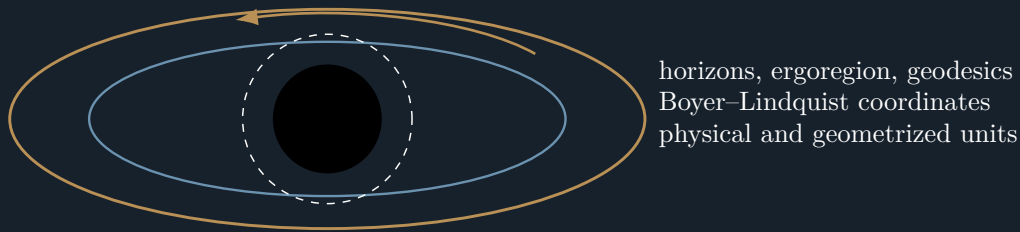


GENERAL RELATIVITY • EINSTEIN–MAXWELL

Mass, Charge, and Angular Momentum

From Einstein–Maxwell to the Kerr–Newman Metric

Theoretical foundations and a rigorous guide to the black-hole simulator



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Theoretical document for the Kerr–Newman Black Hole Simulator

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Purpose and logical strategy

The aim of document is to show how the global quantities that characterise a stationary black hole lead to the geometry actually used by the simulator. The route therefore does not begin by stating the Kerr–Newman metric, it begins with the relativistic notions of mass, charge, and angular momentum, constructs the Einstein–Maxwell system, and follows its consequences through to Boyer–Lindquist coordinates and the numerically integrated geodesic equations.

The expository strategy follows that used in *From Maxwell to the Electromagnetic Field Tensor*: first the problem is stated, then the assumptions are made explicit, the result is then derived, and finally it is subjected to a physical check or to a known limit. In this way, every object that is introduced must answer a definite question. Furthermore I will sometimes repeat the same point in different ways to ensure clarity and to allow everyone to start reading at any point of the document, without forcing them to read the entire document sequentially.

The logical thread passes through five levels:

1. Identify what mass M , charge Q , and angular momentum J represent in Relativity;
2. Understand why the electromagnetic field requires a stress–energy tensor and how it enters Einstein’s equations;
3. Derive Reissner–Nordström and use this solution to distinguish the outer horizon, the inner horizon, and a curvature singularity;
4. Move from spherical symmetry to rotation, construct Boyer–Lindquist coordinates, and read in the Kerr–Newman metric the objects displayed by the simulator;
5. Using all the tools available to understand the relation between the geometry, initial conditions, Christoffel symbols, geodesics, and numerical integration.

The Reissner–Nordström treatment is developed directly from the Einstein–Maxwell equations. For Kerr and Kerr–Newman, however, three conceptually distinct operations must be kept separate: generating or finding a solution, verifying it in the field equations, and rewriting it in a useful coordinate chart. The Newman–Janis method belongs to the first operation; the passage to Boyer–Lindquist coordinates belongs to the third. Keeping these operations distinct prevents a coordinate transformation from being presented as though it produced a new geometry.

The conventions use the signature $(-+++)$, as does the simulator code. Greek indices take the values $0, 1, 2, 3$; repeated indices are summed; and the connection is the Levi–Civita connection. Square brackets denote antisymmetrisation. Except when SI units are explicitly restored, geometrized units are used,

$$G = c = 1. \tag{1}$$

The electromagnetic sector uses geometrized Gaussian units; SI formulas appear only where SI units are explicitly restored. In these units, mass, geometrized charge, and the spin parameter have the dimension of length, whereas geometrized angular momentum has the dimension of length squared. The time coordinate t also has the dimension of length; on returning to SI units, one writes $t = ct_{\text{SI}}$.

Notation

Q normally denotes the *geometrized* charge; when necessary, Q_{SI} and Q_g distinguish coulombs from the equivalent length. The parameter $a = J/M$ is angular momentum per unit mass, not J . Since the solver sets $M = 1$, the numerical value of a coincides with the dimensionless spin $a_* = a/M$.

Contents

Purpose and logical strategy	i
1 Mass, Charge and Spin in Relativity	1
1.1 Three global quantities, not three simple “labels”	1
1.2 Mass: from the particle invariant to ADM mass	1
1.3 Charge: conserved current and flux at infinity	5
1.4 Angular momentum and the Kerr parameter	8
1.5 Dimensionless parameters and extremality	9
1.6 Conversions implemented in the simulator	10
2 Covariant electromagnetism in curved spacetime	11
2.1 The problem: taking Maxwell theory into curved spacetime	11
2.1.1 Why replacing partial derivatives by covariant derivatives does not change the electromagnetic field	11
2.2 Charge conservation as a consequence	13
2.3 Why the electromagnetic field must gravitate	14
2.3.1 Electromagnetic action and metric variation	14
2.3.2 First check: positive energy density	14
2.3.3 Second check: vanishing trace	15
2.4 The coupled Einstein–Maxwell system	15
3 Derivation of the Reissner–Nordström metric	16
3.1 Strategy of the derivation	16
3.2 Geometrical assumptions	16
3.2.1 Killing vectors	17
3.2.2 Electromagnetic ansatz	19
3.3 First step: solving Maxwell’s equation	20
3.4 Second step: constructing $T^\mu{}_\nu$	23
3.5 Third step: Einstein’s equations	23
3.5.1 Determination of $g_{tt}g_{rr}$	24
3.5.2 Integration of the radial equation	24
3.5.3 Interpretation through the mass function	25
3.6 Final solution and potential	25
3.6.1 Three immediate checks	25
3.7 Coordinate singularities and curvature singularities	26

4	The two Reissner–Nordström horizons	27
4.1	The geometrical question	27
4.2	Roots and factorization	27
4.3	The outer horizon r_+	27
4.4	Tortoise coordinate and extension across r_+	28
4.4.1	Future direction inside the outer horizon	29
4.4.2	Kruskal-type coordinates and surface gravity	30
4.5	The inner horizon r_- : the Cauchy horizon	30
4.6	The three regimes	31
4.6.1	Subextremal: $M^2 > Q^2$	31
4.6.2	Extremal: $M^2 = Q^2$	31
4.6.3	Superextremal: $M^2 < Q^2$	31
4.7	Photon orbit in Reissner–Nordström	32
4.8	Physical interpretation: collapse and eternal solutions	32
5	From spherical symmetry to rotation	34
5.1	Staticity, stationarity, and axisymmetry	34
5.2	Why rotating the coordinates is not enough	35
5.3	One route to the solution: Kerr–Schild and Newman–Janis	36
6	Boyer–Lindquist coordinates	38
6.1	From advanced coordinates to Boyer–Lindquist coordinates	38
6.2	Geometrical meaning: oblate spheroidal coordinates	40
6.3	Why they are so useful	41
7	The Kerr–Newman metric and its limiting cases	42
7.1	What the solution must contain	42
7.2	Compact form	42
7.3	Expanded form: the one implemented in the code	43
7.4	Inverse metric and determinant	43
7.5	The Kerr–Newman electromagnetic field	44
7.6	Checks in the four limits	45
7.7	SI units in the function Δ	47
8	Horizons, ergoregion, and the rotating singularity	48
8.1	The horizons: $\Delta = 0$	48
8.2	Static limit: $g_{tt} = 0$	50
8.3	Ergoregion and inertial-frame dragging	52
8.4	Ring singularity	54

8.5	Area, surface gravity, and horizon potential	55
8.6	Negative energy and the Penrose process	57
9	Kerr–Newman geodesics	59
9.1	Lagrangian and normalization	59
9.2	Energy and axial angular momentum	60
9.3	Carter constant and separability	61
9.4	Why the equatorial plane remains equatorial	61
9.5	Construction of the initial radial velocity	63
10	Distinguished orbits and physical presets	64
10.1	Schwarzschild: photon sphere and ISCO	64
10.2	Kerr: prograde and retrograde orbits	67
10.3	What the Web Demo does and does not show	68
10.4	Simulation presets	69
11	From the tensor to the numerical code	70
11.1	From the continuous problem to the numerical state	70
11.2	Construction of the metric	70
11.3	Christoffel symbols	70
11.4	First-order system	71
11.5	Fourth-order Runge–Kutta method	71
11.6	Actual numerical parameters	72
11.7	Conservation checks	72
11.8	Stopping criteria	72
12	Physical guide to the simulator interface	73
12.1	Black-hole parameters	73
12.2	Initial conditions	73
12.3	Properties calculated in real time	74
12.4	Visualization: reading the plot correctly	74
12.5	What is not yet simulated	74
13	Essential algebraic checks	75
13.1	From the compact squares to the components	75
13.2	Determinant of the t – ϕ block	75
13.3	Why the horizon does not coincide with $g_{tt} = 0$	75
A	Quick reference for units	76

B	Map between theory and implementation	77
C	Minimal glossary	78
	Conclusion	79

Mass, Charge and Spin in Relativity

1.1 Three global quantities, not three simple “labels”

The first problem is to understand the parameters that appear in the metric. A stationary, asymptotically flat, isolated black hole is not described by arbitrarily assigning a mass density inside a sphere. For an external observer, the relevant quantities are instead global charges which are associated with the entire solution and measurable through the behaviour of fields at large distance or through integrals over closed surfaces.

This distinction is essential. A density depends on the spacetime point and on the observer; a global charge is obtained by integrating a conservation law and is not tied to a particular local parametrisation of the interior. In the Kerr–Newman case, the three charges are:

1. Total gravitational energy, expressed as the ADM mass M ;
2. Total electric charge Q ;
3. Total angular momentum J about the symmetry axis.

Under the hypotheses of the electrovacuum uniqueness theorems, these quantities characterise the Kerr–Newman family. This is the physical content of the expression “black holes have no hair”, which must be understood with the theorem’s precise mathematical hypotheses, not as a universal claim about every theory, dynamical configuration, or external matter distribution [1, 11].

The hierarchy of solutions is

Solution	Parameters	Symmetry	External field
Schwarzschild	M	static, spherical	vacuum
Reissner–Nordström	M, Q	static, spherical	electrovacuum
Kerr	M, J	stationary, axisymmetric	vacuum
Kerr–Newman	M, Q, J	stationary, axisymmetric	electrovacuum

1.2 Mass: from the particle invariant to ADM mass

For a free particle in special relativity, the four-momentum is

$$p^\mu = mu^\mu, \quad u^\mu = \frac{dx^\mu}{d\tau}, \quad u^\mu u_\mu = -c^2, \quad (1.1)$$

where τ is proper time. The rest mass is therefore encoded in the invariant

$$p^\mu p_\mu = -m^2 c^2. \quad (1.2)$$

In an inertial frame, with $p^\mu = (E/c, \mathbf{p})$ and signature $(-+++)$, this becomes

$$-\frac{E^2}{c^2} + \mathbf{p}^2 = -m^2 c^2, \quad \implies \quad E^2 = \mathbf{p}^2 c^2 + m^2 c^4. \quad (1.3)$$

This definition refers to a single particle: its four-momentum $p^\mu(p)$ is a vector in the tangent space $T_p\mathcal{M}$ at the spacetime point p . In a curved spacetime, however, a momentum $p^\mu(q)$ defined at another point q belongs to a different tangent space,

$$p^\mu(p) \in T_p\mathcal{M}, \quad p^\mu(q) \in T_q\mathcal{M}, \quad T_p\mathcal{M} \neq T_q\mathcal{M}. \quad (1.4)$$

There is therefore no canonical sum $p^\mu(p) + p^\mu(q)$. To compare the two vectors, one would first have to parallel-transport one of them to the other point. In the presence of curvature, the result generally depends on the path, as expressed by

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu}V^\sigma. \quad (1.5)$$

The invariant definition of the mass of a particle thus does not automatically extend to the notion of the total mass of an entire spacetime.

There is a second difficulty. General relativity has no local, covariant, and unique density of gravitational energy. At every point p , one may choose locally inertial coordinates such that

$$g_{\mu\nu}(p) = \eta_{\mu\nu}, \quad \partial_\alpha g_{\mu\nu}(p) = 0, \quad \Gamma^\rho_{\mu\nu}(p) = 0. \quad (1.6)$$

A quantity constructed only from Christoffel symbols could therefore be made to vanish in one coordinate system without representing the absence of physical gravity. Curvature, which contains second derivatives of the metric, cannot be eliminated in this way; nevertheless, it does not provide a universal local gravitational energy–momentum tensor. Total gravitational energy must instead be defined globally, or at least quasi-locally.

For an isolated, asymptotically flat spacetime, total energy can be defined from the asymptotic geometry. This is the ADM mass.

The spatial metric and the asymptotic perturbation

Let $\mathcal{S}_t$ be a spacelike hypersurface defined by $t = \text{constant}$, and let n^μ be its future-directed unit normal:

$$n^\mu n_\mu = -1. \quad (1.7)$$

The metric induced on $\mathcal{S}_t$ is

$$\gamma_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu.^1 \quad (1.8)$$

It is indeed purely spatial, because

$$\gamma_{\mu\nu}n^\nu = g_{\mu\nu}n^\nu + n_\mu(n_\nu n^\nu) = n_\mu - n_\mu = 0. \quad (1.9)$$

If X^μ is tangent to $\mathcal{S}_t$, then $n_\mu X^\mu = 0$ and

$$\gamma_{\mu\nu}X^\mu X^\nu = g_{\mu\nu}X^\mu X^\nu > 0. \quad (1.10)$$

In spatial coordinates x^i on $\mathcal{S}_t$, with $i, j = 1, 2, 3$, one therefore writes

$$d\ell^2 = \gamma_{ij}dx^i dx^j. \quad (1.11)$$

¹Here $\gamma_{\mu\nu}$ is a covariant tensor of type $(0, 2)$, written with four-dimensional spacetime indices. Its matrix has rank three, since it annihilates the timelike unit normal: $\gamma_{\mu\nu}n^\nu = 0$. Thus it is degenerate as a bilinear form on the full spacetime, while its restriction to vectors tangent to the spacelike hypersurface Σ is the non-degenerate three-dimensional induced metric. Its intrinsic components on Σ are

$$\gamma_{ij} = g_{\mu\nu}e_i^\mu e_j^\nu,$$

where $e_i^\mu = \partial X^\mu / \partial y^i$ are tangent vectors to Σ . The plus sign assumes the signature $(-+++)$ and a unit timelike normal $n^\mu n_\mu = -1$.

Asymptotic flatness means that sufficiently far from the source one can choose asymptotically Cartesian coordinates in which γ_{ij} approaches the Euclidean metric

$$\delta_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad d\ell_{\text{flat}}^2 = \delta_{ij} dx^i dx^j = dx^2 + dy^2 + dz^2. \quad (1.12)$$

The tensor

$$\boxed{h_{ij} := \gamma_{ij} - \delta_{ij}} \iff \boxed{\gamma_{ij} = \delta_{ij} + h_{ij}} \quad (1.13)$$

measures the deviation of the physical spatial geometry from Euclidean geometry.

No smallness condition is imposed on h_{ij} everywhere: near a black hole, the geometry can differ drastically from flat space. The required condition is only that, as $r \rightarrow \infty$,

$$h_{ij} = O(r^{-1}), \quad \partial_k h_{ij} = O(r^{-2}). \quad (1.14)$$

Spatial indices in the asymptotic region are raised and lowered with δ_{ij} ; in particular,

$$h_{jj} := \delta^{jk} h_{jk} = h_{11} + h_{22} + h_{33} \quad (1.15)$$

is the trace of h_{ij} .

Meaning of γ_{ij} , δ_{ij} , and h_{ij}

γ_{ij} is the physical spatial metric on a constant-time hypersurface. δ_{ij} is the flat Euclidean metric used as the asymptotic reference geometry. Their difference h_{ij} contains the asymptotic gravitational information from which the total mass is extracted.

Origin of the ADM surface term

The combination appearing in the ADM formula arises from the scalar curvature of the spatial metric. To first order in h_{ij} , the three-dimensional Christoffel symbols are

$${}^{(3)}\Gamma^k_{ij} = \frac{1}{2} \delta^{k\ell} (\partial_i h_{j\ell} + \partial_j h_{i\ell} - \partial_\ell h_{ij}) + O(h \partial h). \quad (1.16)$$

Substituting this expression into the definition of the Ricci tensor and contracting indices gives

$${}^{(3)}R = \partial_i \partial_j h_{ij} - \partial_i \partial_i h_{jj} + O(h \partial^2 h + (\partial h)^2). \quad (1.17)$$

The linear term is a divergence:

$${}^{(3)}R^{(1)} = \partial_i (\partial_j h_{ij} - \partial_i h_{jj}). \quad (1.18)$$

The divergence theorem applied to a large spatial region $\mathcal{B}_r$ bounded by a sphere S_r then yields

$$\int_{\mathcal{B}_r} {}^{(3)}R^{(1)} d^3x = \int_{S_r} (\partial_j h_{ij} - \partial_i h_{jj}) n^i dS, \quad (1.19)$$

where $n^i = x^i/r$ is the outward unit normal and $dS = r^2 d\Omega$ is the area element of the sphere.

Equation (1.19) explains the geometrical origin of the ADM integrand. In the full Hamiltonian formulation of general relativity, this boundary term is required for the generator of asymptotic

time translations to be well defined. On solutions of Einstein's constraint equations, the volume contribution vanishes and the total energy is the contribution at spatial infinity:

$$\boxed{M_{\text{ADM}}^2 = \frac{1}{16\pi} \lim_{r \rightarrow \infty} \int_{S_r} (\partial_j h_{ij} - \partial_i h_{jj}) n^i dS} \quad (G = c = 1). \quad (1.20)$$

The fall-off conditions (1.14) ensure that this limit is finite: $\partial h = O(r^{-2})$, while the area of S_r grows as r^2 .

Where the parameter M comes from

In the weak-field limit and for slowly moving particles, the temporal component of the metric takes the form

$$g_{tt} = - \left(1 + \frac{2\Phi}{c^2} \right) + O\left(\frac{\Phi^2}{c^4} \right), \quad (1.21)$$

where Φ is the Newtonian potential. Indeed, with $x^0 = ct$, the spatial geodesic equation reduces to

$$\frac{d^2 x^i}{dt^2} \simeq -c^2 \Gamma^i_{00} = -c^2 \left(-\frac{1}{2} \delta^{ij} \partial_j g_{00} \right) = -\partial^i \Phi. \quad (1.22)$$

Thus the geodesic equation reproduces Newton's law.

For an isolated source,

$$\Phi(r) = -\frac{GM_{\text{SI}}}{r}, \quad (1.23)$$

and hence

$$g_{tt} = -1 + \frac{2GM_{\text{SI}}}{c^2 r} + O(r^{-2}). \quad (1.24)$$

Defining the geometrized mass

$$\boxed{M_g = \frac{GM_{\text{SI}}}{c^2}}, \quad (1.25)$$

one obtains, in geometrized units,

$$\boxed{g_{tt} = -1 + \frac{2M}{r} + O(r^{-2})}. \quad (1.26)$$

²The factor $1/(16\pi)$ fixes the normalization of the ADM mass. To see this explicitly, consider an asymptotically Schwarzschild geometry. In asymptotically Cartesian coordinates, the spatial metric behaves as

$$\gamma_{ij} = \delta_{ij} + \frac{2M}{r} \delta_{ij} + O(r^{-2}),$$

so that

$$h_{ij} \equiv \gamma_{ij} - \delta_{ij} = \frac{2M}{r} \delta_{ij} + O(r^{-2}).$$

Using $\partial_i(r^{-1}) = -x_i/r^3$, one obtains

$$\partial_j h_{ij} - \partial_i h_{jj} = \frac{4M x_i}{r^3} + O(r^{-3}).$$

Since the outward surface element on a large sphere is $dS^i = (x^i/r) r^2 d\Omega$, the surface integral becomes

$$\lim_{r \rightarrow \infty} \oint_{S_r} (\partial_j h_{ij} - \partial_i h_{jj}) dS^i = 4M \int_{S^2} d\Omega = 16\pi M.$$

The prefactor $1/(16\pi)$ therefore ensures that the ADM expression returns precisely the Schwarzschild mass M . More fundamentally, this normalization is inherited from the Einstein–Hilbert action, whose gravitational prefactor is $1/(16\pi G)$. In geometrized units, $G = c = 1$, this reduces to $1/(16\pi)$; with physical units restored, the ADM mass formula carries the prefactor $c^2/(16\pi G)$.

The coefficient M of the $1/r$ term is precisely the ADM mass. Conceptually, M is not assigned as a density at the centre: it is read from the asymptotic deviation of the geometry from Minkowski spacetime.

This can be checked directly for Schwarzschild. In isotropic coordinates ρ , the spatial metric is

$$\gamma_{ij} = \left(1 + \frac{M}{2\rho}\right)^4 \delta_{ij} = \left(1 + \frac{2M}{\rho} + O(\rho^{-2})\right) \delta_{ij}. \quad (1.27)$$

Therefore,

$$h_{ij} = \frac{2M}{\rho} \delta_{ij} + O(\rho^{-2}). \quad (1.28)$$

At leading order,

$$\partial_j h_{ij} = -\frac{2M x_i}{\rho^3}, \quad \partial_i h_{jj} = -\frac{6M x_i}{\rho^3}, \quad (1.29)$$

and thus

$$(\partial_j h_{ij} - \partial_i h_{jj}) n^i = \frac{4M}{\rho^2}. \quad (1.30)$$

Since $\int_{S_\rho} dS = 4\pi\rho^2$, (1.20) gives

$$M_{\text{ADM}} = \frac{1}{16\pi} \lim_{\rho \rightarrow \infty} \int_{S_\rho} \frac{4M}{\rho^2} dS = M. \quad (1.31)$$

The parameter M in the Schwarzschild metric, and likewise in Reissner–Nordström and Kerr–Newman, is therefore the ADM mass of the spacetime. In geometrized units it has the dimension of length:

$$[M] = [M_g] = \text{length}. \quad (1.32)$$

The Schwarzschild radius is

$$r_S = 2M_g = \frac{2GM_{\text{SI}}}{c^2}. \quad (1.33)$$

Consequently, ratios such as r/M are dimensionless. Setting $M = 1$ in the simulator means using the geometrized mass as the unit of length and time, not assigning the black hole a mass of one kilogram.

1.3 Charge: conserved current and flux at infinity

Unlike mass³, electric charge begins with a genuine local density. Let ρ_0 be the charge density measured in the instantaneous rest frame of the charged matter. In that frame, no charge flows spatially, so the current has the form

$$J_{\text{rest}}^{\hat{\mu}} = (\rho_0, 0, 0, 0) \quad (c = 1). \quad (1.34)$$

Since the four-velocity in the rest frame is

$$u_{\text{rest}}^{\hat{\mu}} = (1, 0, 0, 0), \quad (1.35)$$

Lorentz covariance implies that the four-current is

$$\boxed{J^\mu = \rho_0 u^\mu}. \quad (1.36)$$

³Before reading this section and the next chapter I strongly suggest to go pay a visit to my other document about the Electromagnetic Field Tensor located in the Notes section of my site. Although I did not introduced the general covariant form, for now let us assume that it holds true even with the covariant derivative. In chapter 2 I will clarify this further.

Thus the temporal component of J^μ represents charge density in a chosen frame, while the spatial components represent charge flux.

The local conservation law is

$$\boxed{\nabla_\mu J^\mu = 0.} \quad (1.37)$$

To see its coordinate content, expand the covariant divergence:

$$\nabla_\mu J^\mu = \partial_\mu J^\mu + \Gamma^\mu_{\mu\nu} J^\nu. \quad (1.38)$$

Using

$$\Gamma^\mu_{\mu\nu} = \frac{1}{2} g^{\alpha\beta} \partial_\nu g_{\alpha\beta} = \partial_\nu \ln \sqrt{-g}, \quad (1.39)$$

where $g = \det(g_{\mu\nu})$, one obtains

$$\boxed{\nabla_\mu J^\mu = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} J^\mu).} \quad (1.40)$$

Equation (1.37) is therefore the curved-spacetime generalization of the familiar continuity equation.

Let $\mathcal{V}$ be a four-dimensional region bounded by two spacelike hypersurfaces Σ_{t_1} and Σ_{t_2} , together with a timelike lateral boundary $\mathcal{T}$. Applying the covariant divergence theorem gives

$$0 = \int_{\mathcal{V}} \nabla_\mu J^\mu \sqrt{-g} d^4x = \int_{\partial\mathcal{V}} J^\mu d\Sigma_\mu. \quad (1.41)$$

The boundary integral separates into the charge on the final hypersurface, the charge on the initial hypersurface, and the current flowing through the lateral boundary. Thus,

$$Q_{\Sigma_{t_2}} - Q_{\Sigma_{t_1}} = - \int_{\mathcal{T}} J^\mu d\Sigma_\mu. \quad (1.42)$$

If no charge crosses the lateral boundary, then

$$Q_{\Sigma_{t_2}} = Q_{\Sigma_{t_1}}. \quad (1.43)$$

The total charge is therefore a conserved global quantity.

We can express the charge contained in a spacelike hypersurface directly in tensor notation. Let n^μ be the future-directed unit normal to the hypersurface Σ , normalized according to

$$n^\mu n_\mu = -1, \quad (1.44)$$

and let

$$d\Sigma = \sqrt{\gamma} d^3y \quad (1.45)$$

be its proper three-dimensional volume element, where $\gamma = \det(\gamma_{ij})$ is the determinant of the induced spatial metric. The charge density measured by the observers whose four-velocity is n^μ is

$$\rho_{(n)} = -J^\mu n_\mu. \quad (1.46)$$

The minus sign follows from the signature $(-+++)$ and ensures that $\rho_{(n)} = \rho_0$ when the observers are comoving with the charged matter. Indeed, if $n^\mu = u^\mu$, then

$$-J^\mu n_\mu = -\rho_0 u^\mu u_\mu = \rho_0. \quad (1.47)$$

The total charge on Σ is therefore

$$\boxed{Q_\Sigma = - \int_\Sigma J^\mu n_\mu d\Sigma.} \quad (1.48)$$

This expression is completely covariant: $J^\mu n_\mu$ is a scalar, while $d\Sigma$ is the invariant proper-volume element on the three-dimensional hypersurface.

Maxwell's equations in geometrized Gaussian conventions are

$$\boxed{\nabla_{[\lambda} F_{\mu\nu]} = 0, \quad \nabla_\mu F^{\mu\nu} = 4\pi J^\nu.} \quad (1.49)$$

The first equation is the tensorial form of the homogeneous Maxwell equations, while the second contains Gauss's law and the Ampère–Maxwell law.

To extract Gauss's law, define the electric field measured by the observers with four-velocity n^μ as

$$E^\mu = -F^{\mu\nu} n_\nu. \quad (1.50)$$

Because $F^{\mu\nu}$ is antisymmetric,

$$E^\mu n_\mu = 0, \quad (1.51)$$

so E^μ is tangent to the spatial hypersurface Σ . Projecting the inhomogeneous Maxwell equation along n^μ gives the spatial Gauss constraint

$$\boxed{D_\mu E^\mu = 4\pi \rho_{(n)},} \quad (1.52)$$

where D_μ is the covariant derivative compatible with the induced spatial metric $\gamma_{\mu\nu}$.

In an electrovacuum region, there is no material current:

$$J^\mu = 0. \quad (1.53)$$

Consequently,

$$\rho_{(n)} = -J^\mu n_\mu = 0, \quad (1.54)$$

and Gauss's law reduces to

$$D_\mu E^\mu = 0. \quad (1.55)$$

Consider the spatial shell $\mathcal{A}$ bounded by two spheres S_1 and S_2 , both enclosing the sources. Let s^μ denote the outward unit normal to each sphere within the spatial hypersurface, and let dA be the proper area element induced on the sphere. Applying the three-dimensional divergence theorem gives

$$0 = \int_{\mathcal{A}} D_\mu E^\mu d\Sigma = \int_{S_2} E^\mu s_\mu dA - \int_{S_1} E^\mu s_\mu dA. \quad (1.56)$$

Hence,

$$\int_{S_2} E^\mu s_\mu dA = \int_{S_1} E^\mu s_\mu dA. \quad (1.57)$$

The electromagnetic flux is independent of the sphere chosen, provided that no material charge lies between the two spheres. The total charge can consequently be read at infinity:

$$\boxed{Q = \frac{1}{4\pi} \int_{S_\infty^2} E^\mu s_\mu dA.} \quad (1.58)$$

Why charge can be measured at infinity

The local conservation law says that charge cannot disappear inside a spacetime region. Maxwell's equation then converts the total enclosed charge into a flux of the electromagnetic field. In an electrovacuum region, the same flux crosses every sphere surrounding the source; this is why Q is a global asymptotic charge.

The charge entering the metric must have the dimension of length. This conversion follows directly by comparing the Reissner–Nordström metric in SI units,

$$f(r) = 1 - \frac{2GM_{\text{SI}}}{c^2 r} + \frac{GQ_{\text{SI}}^2}{4\pi\epsilon_0 c^4 r^2}, \quad (1.59)$$

with its geometrized form,

$$f(r) = 1 - \frac{2M_g}{r} + \frac{Q_g^2}{r^2}. \quad (1.60)$$

Equating the coefficients gives

$$Q_g = \sqrt{\frac{G}{4\pi\epsilon_0}} \frac{Q_{\text{SI}}}{c}. \quad (1.61)$$

The sign of Q changes the orientation of the electromagnetic field and therefore the Lorentz force acting on a charged test particle. However, the electromagnetic stress–energy tensor is quadratic in F :

$$T_{\mu\nu}^{\text{EM}} = \frac{1}{4\pi} \left(F_{\mu\alpha} F_{\nu}{}^{\alpha} - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right). \quad (1.62)$$

Under

$$F_{\mu\nu} \longrightarrow -F_{\mu\nu}, \quad Q \longrightarrow -Q, \quad (1.63)$$

one has

$$T_{\mu\nu}^{\text{EM}} \longrightarrow T_{\mu\nu}^{\text{EM}}. \quad (1.64)$$

The geometry therefore depends on Q^2 , as is evident in (1.60). The spacetimes generated by $+Q$ and $-Q$ are isometric, although their electromagnetic fields have opposite orientation and act differently on charged particles.

1.4 Angular momentum and the Kerr parameter

For a particle, relativistic spin can be encoded in the Pauli–Lubanski vector. For a stationary, axisymmetric spacetime, J is instead the global charge associated with the axial Killing vector field

$$\eta^\mu = \left(\frac{\partial}{\partial \phi} \right)^\mu.{}^4 \quad (1.65)$$

Because η^μ is a Killing vector field, it satisfies the Killing equation

$$\nabla_{(\mu} \eta_{\nu)} = \frac{1}{2} (\nabla_\mu \eta_\nu + \nabla_\nu \eta_\mu) = 0. \quad (1.66)$$

Consequently, its covariant derivative is antisymmetric:

$$\nabla_\mu \eta_\nu = -\nabla_\nu \eta_\mu. \quad (1.67)$$

⁴A Killing vector is a vector field ζ^μ that generates an isometry of the spacetime metric, namely

$$\mathcal{L}_\zeta g_{\mu\nu} = 0 \quad \Longleftrightarrow \quad \nabla_{(\mu} \zeta_{\nu)} = 0.$$

In an axisymmetric spacetime, rotations about the symmetry axis are generated by

$$\eta^\mu = \left(\frac{\partial}{\partial \phi} \right)^\mu,$$

whose components in adapted coordinates (t, r, θ, ϕ) are $\eta^\mu = (0, 0, 0, 1)$. Since an axisymmetric metric is independent of ϕ ,

$$\mathcal{L}_\eta g_{\mu\nu} = \partial_\phi g_{\mu\nu} = 0,$$

so η^μ is a Killing vector. The associated conserved quantity for a test particle is $p_\mu \eta^\mu = p_\phi$, i.e. its angular momentum about the symmetry axis.

Let S_∞^2 be a two-sphere at spatial infinity. Let n^μ be the future-directed timelike unit normal to a spacelike hypersurface Σ , and let s^μ be the outward-directed spacelike unit normal to S_∞^2 within Σ . These vectors satisfy

$$n^\mu n_\mu = -1, \quad s^\mu s_\mu = 1, \quad n^\mu s_\mu = 0. \quad (1.68)$$

The oriented antisymmetric surface element of S_∞^2 is

$$dS_{\mu\nu} = 2n_{[\mu}s_{\nu]} dA = (n_\mu s_\nu - n_\nu s_\mu) dA, \quad (1.69)$$

where dA is the proper area element induced on the two-sphere.

With the usual orientation conventions, the total angular momentum at infinity is represented by the Komar integral

$$J = \frac{1}{16\pi} \int_{S_\infty^2} \nabla^\mu \eta^\nu dS_{\mu\nu}. \quad (1.70)$$

Since $\nabla^\mu \eta^\nu$ is antisymmetric, only the antisymmetric surface element contributes to the integral. Equivalently,

$$\nabla^\mu \eta^\nu dS_{\mu\nu} = 2n_\mu s_\nu \nabla^\mu \eta^\nu dA. \quad (1.71)$$

The overall sign may be reversed if the orientation of the two-sphere or the convention for $dS_{\mu\nu}$ is reversed. Here too, the parameter is read from the asymptotic geometry. In the Kerr–Newman family one introduces

$$a = \frac{J}{M} \quad (1.72)$$

in geometrized units. In SI units:

$$J_g = \frac{GJ_{\text{SI}}}{c^3}, \quad a = \frac{J_g}{M_g} = \frac{J_{\text{SI}}}{M_{\text{SI}}c}. \quad (1.73)$$

The dimensionless parameter is

$$a_* = \frac{a}{M_g} = \frac{cJ_{\text{SI}}}{GM_{\text{SI}}^2}. \quad (1.74)$$

This identification can be checked without using the full integral.

1.5 Dimensionless parameters and extremality

The simulator must compare charge and spin with the scale fixed by the mass. For this purpose, define

$$q_* = \frac{Q_g}{M_g} = \frac{Q_{\text{SI}}}{M_{\text{SI}}\sqrt{4\pi\epsilon_0 G}}. \quad (1.75)$$

The chapter devoted to the metric will show that the horizons are the roots of the polynomial

$$\Delta(r) = r^2 - 2Mr + a^2 + Q^2. \quad (1.76)$$

Its discriminant is $4(M^2 - a^2 - Q^2)$. A real Kerr–Newman horizon therefore exists if and only if

$$a_*^2 + q_*^2 \leq 1. \quad (1.77)$$

Equality identifies the extremal case; values greater than one are superextremal and do not describe a black hole with a horizon in the classical Kerr–Newman family. This is a statement about the exact solution, not a general proof of cosmic censorship or an automatic prediction of the outcome of a dynamical process.

For intuition

Mass, charge, and spin compete in the radial function Δ . Mass favours horizon formation, whereas charge and rotation reduce the separation between the outer and inner horizons. At extremality, the two merge; beyond extremality, the root becomes imaginary and the singularity is no longer screened.

1.6 Conversions implemented in the simulator

The code accepts M_{SI} in solar masses, Q_{SI} in coulombs, and J_{SI} in $\text{kg m}^2/\text{s}$. It then converts them as follows:

$$M_g = \frac{GM_{\text{SI}}}{c^2}, \quad Q_g = \sqrt{\frac{G}{4\pi\epsilon_0}} \frac{Q_{\text{SI}}}{c^2}, \quad J_g = \frac{GJ_{\text{SI}}}{c^3}, \quad a = \frac{J_g}{M_g}. \quad (1.78)$$

Finally, it divides every length by M_g , so that the solver always works with

$$M = 1, \quad a = a_*, \quad Q = q_*. \quad (1.79)$$

For $M_{\text{SI}} = 10M_{\odot}$, $M_g \simeq 14.7667 \text{ km}$. The Kerr parameter is $a = 0.7M_g \simeq 10.3367 \text{ km}$, whereas $J_{\text{SI}} \simeq 6.162 \times 10^{43} \text{ kg m}^2/\text{s}$. The horizons are $r_+ \simeq 25.3122 \text{ km}$ and $r_- \simeq 4.2212 \text{ km}$. The solver simply sees $M = 1$, $a = 0.7$, $r_+ = 1 + \sqrt{1 - 0.7^2}$.

Covariant electromagnetism in curved spacetime

2.1 The problem: taking Maxwell theory into curved spacetime

The document *From Maxwell to the Electromagnetic Field Tensor* showed that, in special relativity, charge density and current density are the components of a four-vector J^μ , whereas the electric and magnetic fields are observer-dependent decompositions of a single antisymmetric tensor $F_{\mu\nu}$. This result is not merely compact notation: it is the structure required by the Lorentz covariance of the Lorentz force.

The passage to general relativity therefore poses a well-defined question: how should the same laws be written when no global inertial frame exists and the metric varies from point to point? The answer must satisfy three requirements:

1. Be tensorial under general coordinate transformations;
2. Reduce to the special-relativistic Maxwell equations in a locally inertial frame;
3. Preserve charge conservation and gauge invariance.

The minimal-coupling prescription¹ suggests replacing ordinary derivatives with covariant derivatives. In the electromagnetic case, however, it is important to verify explicitly why this replacement does not introduce unwanted connection terms.

2.1.1 Why replacing partial derivatives by covariant derivatives does not change the electromagnetic field

The minimal-coupling prescription replaces ordinary derivatives by covariant derivatives when passing from special relativity to curved spacetime. For a covector A_μ , the covariant derivative is

$$\nabla_\mu A_\nu = \partial_\mu A_\nu - \Gamma^\rho_{\mu\nu} A_\rho. \quad (2.1)$$

At first sight, replacing $\partial_\mu A_\nu$ by $\nabla_\mu A_\nu$ seems to introduce additional Christoffel-symbol terms into the electromagnetic field tensor. However, the field is defined by the antisymmetric combination

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (2.2)$$

Using covariant derivatives instead gives

$$\begin{aligned} \nabla_\mu A_\nu - \nabla_\nu A_\mu &= \partial_\mu A_\nu - \Gamma^\rho_{\mu\nu} A_\rho - \partial_\nu A_\mu + \Gamma^\rho_{\nu\mu} A_\rho \\ &= \partial_\mu A_\nu - \partial_\nu A_\mu + (\Gamma^\rho_{\nu\mu} - \Gamma^\rho_{\mu\nu}) A_\rho. \end{aligned} \quad (2.3)$$

For the torsion-free Levi-Civita connection,

$$\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu}, \quad (2.4)$$

¹Here, the phrase “minimal-coupling prescription” refers to the standard gravitational rule for transferring a theory from Minkowski spacetime to curved spacetime:

$$\eta_{\mu\nu} \longrightarrow g_{\mu\nu}, \quad \partial_\mu \longrightarrow \nabla_\mu, \quad d^4x \longrightarrow \sqrt{-g} d^4x.$$

It preserves the local form of the special-relativistic theory while introducing the spacetime geometry through the metric and the Levi-Civita connection. In the electromagnetic case, this replacement must nevertheless be checked explicitly; for $F_{\mu\nu}$, the connection terms cancel because the connection is torsion-free and $F_{\mu\nu}$ is antisymmetric.

and therefore the connection terms cancel exactly:

$$\boxed{F_{\mu\nu} = 2\nabla_{[\mu}A_{\nu]} = 2\partial_{[\mu}A_{\nu]} = \partial_{\mu}A_{\nu} - \partial_{\nu}A_{\mu}.} \quad (2.5)$$

Thus, in this particular antisymmetric combination, the covariant derivative produces exactly the same electromagnetic tensor as the ordinary derivative.

The same cancellation occurs in the homogeneous Maxwell equation. Indeed,

$$\nabla_{[\lambda}F_{\mu\nu]} = \partial_{[\lambda}F_{\mu\nu]} - \Gamma^{\rho}_{[\lambda\mu}F_{\rho|\nu]} - \Gamma^{\rho}_{[\lambda\nu}F_{\mu|\rho]}. \quad (2.6)$$

The connection terms vanish because the Levi-Civita symbols are symmetric in their lower indices, whereas the antisymmetrization acts on those same indices. Consequently,

$$\boxed{\nabla_{[\lambda}F_{\mu\nu]} = \partial_{[\lambda}F_{\mu\nu]}.} \quad (2.7)$$

The inhomogeneous Maxwell equation, however, contains a genuine covariant divergence:

$$\nabla_{\mu}F^{\mu\nu} = -4\pi J^{\nu}, \quad (2.8)$$

with the index ordering used in the variation below. We now derive this equation directly from the action.

In geometrized Gaussian units, consider the Einstein–Maxwell action with an external electric current:

$$S[g, A] = \frac{1}{16\pi} \int_{\mathcal{M}} (R - F_{\rho\sigma}F^{\rho\sigma}) \sqrt{-g} d^4x + \int_{\mathcal{M}} J^{\mu} A_{\mu} \sqrt{-g} d^4x. \quad (2.9)$$

When varying with respect to A_{μ} , the metric and the current are held fixed. Since

$$F_{\mu\nu} = \nabla_{\mu}A_{\nu} - \nabla_{\nu}A_{\mu}, \quad (2.10)$$

its variation is

$$\delta F_{\mu\nu} = \nabla_{\mu}\delta A_{\nu} - \nabla_{\nu}\delta A_{\mu}. \quad (2.11)$$

The variation of the electromagnetic part of the action is therefore

$$\begin{aligned} \delta S_{\text{EM}} &= -\frac{1}{8\pi} \int_{\mathcal{M}} F^{\mu\nu} \delta F_{\mu\nu} \sqrt{-g} d^4x \\ &= -\frac{1}{8\pi} \int_{\mathcal{M}} F^{\mu\nu} (\nabla_{\mu}\delta A_{\nu} - \nabla_{\nu}\delta A_{\mu}) \sqrt{-g} d^4x. \end{aligned} \quad (2.12)$$

Using the antisymmetry of $F^{\mu\nu}$,

$$F^{\mu\nu} = -F^{\nu\mu}, \quad (2.13)$$

the two terms in the variation are equal, and hence

$$\delta S_{\text{EM}} = -\frac{1}{4\pi} \int_{\mathcal{M}} F^{\mu\nu} \nabla_{\mu}\delta A_{\nu} \sqrt{-g} d^4x. \quad (2.14)$$

Now use the covariant product rule:

$$F^{\mu\nu} \nabla_{\mu}\delta A_{\nu} = \nabla_{\mu} (F^{\mu\nu} \delta A_{\nu}) - (\nabla_{\mu} F^{\mu\nu}) \delta A_{\nu}. \quad (2.15)$$

Substituting this identity gives

$$\begin{aligned}\delta S_{\text{EM}} = & -\frac{1}{4\pi} \int_{\mathcal{M}} \nabla_{\mu} (F^{\mu\nu} \delta A_{\nu}) \sqrt{-g} d^4x \\ & + \frac{1}{4\pi} \int_{\mathcal{M}} (\nabla_{\mu} F^{\mu\nu}) \delta A_{\nu} \sqrt{-g} d^4x.\end{aligned}\quad (2.16)$$

The first term is a boundary term. Assuming that δA_{μ} vanishes on the boundary, or that the boundary term is cancelled by suitable boundary conditions, it can be discarded. Thus,

$$\delta S_{\text{EM}} = \frac{1}{4\pi} \int_{\mathcal{M}} (\nabla_{\mu} F^{\mu\nu}) \delta A_{\nu} \sqrt{-g} d^4x. \quad (2.17)$$

The variation of the interaction term is

$$\delta S_{\text{int}} = + \int_{\mathcal{M}} J^{\nu} \delta A_{\nu} \sqrt{-g} d^4x. \quad (2.18)$$

Therefore, the total variation is

$$\delta S = \int_{\mathcal{M}} \left[\frac{1}{4\pi} \nabla_{\mu} F^{\mu\nu} + J^{\nu} \right] \delta A_{\nu} \sqrt{-g} d^4x. \quad (2.19)$$

Since δA_{ν} is arbitrary, the Euler–Lagrange equations give

$$\boxed{\nabla_{\mu} F^{\mu\nu} = -4\pi J^{\nu}.} \quad (2.20)$$

The replacement of ordinary derivatives by covariant derivatives is therefore justified in two complementary ways: first, the connection terms cancel in the antisymmetric definition of $F_{\mu\nu}$; second, the covariant Maxwell equation follows directly from the generally covariant Einstein–Maxwell action.

2.2 Charge conservation as a consequence

Current continuity need not be added as an independent equation. Applying ∇_{ν} to (2.20) gives

$$4\pi \nabla_{\nu} J^{\nu} = \nabla_{\nu} \nabla_{\mu} F^{\nu\mu}. \quad (2.21)$$

Because $F^{\nu\mu}$ is antisymmetric, dummy indices can be exchanged in one half of the expression, giving

$$\nabla_{\nu} \nabla_{\mu} F^{\nu\mu} = \frac{1}{2} [\nabla_{\nu}, \nabla_{\mu}] F^{\nu\mu}. \quad (2.22)$$

The commutator produces contractions of the Ricci tensor with F . Every term vanishes because $R_{\mu\nu}$ is symmetric and $F^{\mu\nu}$ is antisymmetric. Therefore

$$\boxed{\nabla_{\mu} J^{\mu} = 0.} \quad (2.23)$$

2.3 Why the electromagnetic field must gravitate

Maxwell's equations determine F , but Einstein's equations require a source $T_{\mu\nu}$. The electromagnetic field carries energy, momentum, and stresses; it therefore cannot be present without contributing to curvature. One must construct a tensor that is

1. Symmetric, because it must couple to $G_{\mu\nu}$;
2. Quadratic in F , since electromagnetic energy is quadratic in the fields;
3. Gauge invariant, hence dependent on F rather than directly on A ;
4. Conserved in electrovacuum, in accordance with the Bianchi identity.

The action supplies all these requirements in a single construction.

2.3.1 Electromagnetic action and metric variation

In geometrized Gaussian units, the free-field action is

$$S_{\text{EM}} = -\frac{1}{16\pi} \int F_{\rho\sigma} F^{\rho\sigma} \sqrt{-g} \, d^4x. \quad (2.24)$$

The stress–energy tensor is defined by the response of the action to a variation of the inverse metric:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{EM}}}{\delta g^{\mu\nu}}. \quad (2.25)$$

During this variation, A_μ , and hence $F_{\mu\nu}$ with both indices lowered, is held fixed. The metric appears in two places:

$$F_{\rho\sigma} F^{\rho\sigma} = g^{\rho\alpha} g^{\sigma\beta} F_{\rho\sigma} F_{\alpha\beta}, \quad \sqrt{-g}. \quad (2.26)$$

The two required variations are

$$\delta(F_{\rho\sigma} F^{\rho\sigma}) = 2F_{\mu\rho} F_\nu{}^\rho \delta g^{\mu\nu}, \quad (2.27)$$

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}. \quad (2.28)$$

Substitution into (2.24) gives

$$\delta S_{\text{EM}} = -\frac{1}{8\pi} \int \sqrt{-g} \left(F_{\mu\rho} F_\nu{}^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma} \right) \delta g^{\mu\nu} \, d^4x. \quad (2.29)$$

Comparison with the definition (2.25) gives

$$\boxed{T_{\mu\nu}^{\text{EM}} = \frac{1}{4\pi} \left(F_{\mu\rho} F_\nu{}^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma} \right)}. \quad (2.30)$$

In SI units, the prefactor $1/4\pi$ is replaced by $1/\mu_0$, with the corresponding normalization of the electric and magnetic components.

2.3.2 First check: positive energy density

In a local orthonormal frame, where the metric takes the Minkowski form at the point under consideration, the component measured by the observer is

$$\boxed{T_{\hat{0}\hat{0}} = \frac{E^2 + B^2}{8\pi} \geq 0}. \quad (2.31)$$

The mixed time–space components encode the Poynting flux, whereas the spatial components form the Maxwell stress tensor. Equation (2.31) checks the sign in (2.30): with signature $(-+++)$, an inconsistent choice would produce a negative energy density.

2.3.3 Second check: vanishing trace

Contracting (2.30) in four dimensions,

$$\begin{aligned} T^\mu{}_\mu &= \frac{1}{4\pi} \left(F^{\mu\rho} F_{\mu\rho} - \frac{1}{4} \delta^\mu{}_\mu F_{\rho\sigma} F^{\rho\sigma} \right) \\ &= \frac{1}{4\pi} \left(F_{\rho\sigma} F^{\rho\sigma} - \frac{4}{4} F_{\rho\sigma} F^{\rho\sigma} \right) = 0. \end{aligned} \quad (2.32)$$

The cancellation is specific to four dimensions. By performing some simplifications the trace of Einstein's equations will imply $R = 0$ in electrovacuum. This does not imply $R_{\mu\nu} = 0$: the Ricci tensor remains nonzero because the field carries energy and stresses.

2.4 The coupled Einstein–Maxwell system

Without a cosmological constant, the electrovacuum action may be written as

$$S[g, A] = \frac{1}{16\pi} \int (R - F_{\rho\sigma} F^{\rho\sigma}) \sqrt{-g} d^4x. \quad (2.33)$$

Variation with respect to A_μ produces Maxwell's equations; variation with respect to $g^{\mu\nu}$ produces Einstein's equations with source (2.30). In the region exterior to material sources, the system is

$$G_{\mu\nu} = 8\pi T_{\mu\nu}^{\text{EM}}, \quad (2.34)$$

$$\nabla_\mu F^{\nu\mu} = 0, \quad (2.35)$$

$$\nabla_{[\lambda} F_{\mu\nu]} = 0. \quad (2.36)$$

Taking the trace of the first equation and using (2.32),

$$-R = 8\pi T^\mu{}_\mu = 0, \quad \implies \quad R = 0. \quad (2.37)$$

In electrovacuum, the gravitational equations can therefore be rewritten as

$$R_{\mu\nu} = 8\pi T_{\mu\nu}^{\text{EM}}. \quad (2.38)$$

The coupling acts in both directions: $F_{\mu\nu}$ curves spacetime through $T_{\mu\nu}^{\text{EM}}$; the metric determines how indices are raised, the Hodge dual, the volume factor $\sqrt{-g}$, and hence the equation satisfied by F . It is precisely this system that produces Reissner–Nordström in the static case and Kerr–Newman in the rotating stationary case.

Electrovacuum does not mean absence of a field

In the exterior region $J^\mu = 0$, but $F_{\mu\nu} \neq 0$ and $T_{\mu\nu}^{\text{EM}} \neq 0$. Reissner–Nordström and Kerr–Newman are therefore not vacuum solutions with $R_{\mu\nu} = 0$; they are Einstein–Maxwell solutions without local material sources. Only by also setting $F_{\mu\nu} = 0$ does one recover the ordinary gravitational vacuum.

Derivation of the Reissner–Nordström metric

3.1 Strategy of the derivation

We seek the simplest solution in which the electromagnetic field genuinely modifies the geometry; the exterior metric of an isolated, static, spherically symmetric configuration carrying total electric charge Q . The calculation is performed in the electrovacuum region outside the material sources, where

$$J^\mu = 0, \quad F_{\mu\nu} \neq 0, \quad T_{\mu\nu}^{\text{EM}} \neq 0. \quad (3.1)$$

Thus, unlike in the Schwarzschild exterior, the gravitational equations are not vacuum equations: the electromagnetic field itself supplies the stress–energy that curves spacetime.

The derivation is organized into four separately verifiable steps:

1. Use the symmetries to restrict the form of the metric and of F ;
2. Solve Maxwell’s equation in electrovacuum;
3. Construct the stress–energy tensor of the resulting field;
4. Integrate Einstein’s equations and fix the constants by asymptotic flatness.

Neither the $1/r^2$ law nor the Q^2/r^2 term in the metric will be assumed in advance. Both will emerge from the equations.

3.2 Geometrical assumptions

We now determine the most general form of the metric compatible with the physical assumptions of the problem. We are interested in the exterior geometry of an isolated, non-rotating charged object where Spherical symmetry is imposed. Spherical symmetry means that, with respect to a distinguished centre, the geometry is invariant under arbitrary spatial rotations: no angular direction is preferred. Equivalently, once the time and radial position are fixed, all angular directions must be physically equivalent. The two-dimensional surfaces obtained by keeping the temporal and radial coordinates fixed are therefore ordinary symmetry two-spheres.

Furthermore, the absence of rotation, $J=0$, is consistent with this symmetry but does not by itself imply it.

Let x denote, for the moment, a generic radial coordinate. Before making any further coordinate choice, the most general line element compatible with spherical symmetry can be written as

$$ds^2 = g_{tt}(t, x) dt^2 + 2g_{tx}(t, x) dt dx + g_{xx}(t, x) dx^2 + R^2(t, x) d\Omega^2, \quad (3.2)$$

where

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (3.3)$$

The form of the angular part follows directly from spherical symmetry. In particular, terms such as

$$dt d\theta, \quad dt d\phi, \quad dx d\theta, \quad dx d\phi$$

cannot appear: any such term would single out a preferred direction tangent to the sphere and would therefore break rotational symmetry. For the same reason, the metric on each symmetry sphere must be proportional to the standard round metric $d\Omega^2$.

The function $R(t, x)$ has a direct geometrical meaning. The area of a symmetry sphere is

$$\mathcal{A} = 4\pi R^2(t, x). \quad (3.4)$$

Therefore, whenever R is a valid radial coordinate, we may use itself as the radial coordinate and define

$$r \equiv R. \quad (3.5)$$

With this choice, r is called the *areal radius*: a sphere labelled by the coordinate r has area $4\pi r^2$. Notice that r is defined through the area of the symmetry spheres and is not, in general, the proper radial distance.

After this coordinate choice, the metric becomes

$$ds^2 = g_{tt}(t, r) dt^2 + 2g_{tr}(t, r) dt dr + g_{rr}(t, r) dr^2 + r^2 d\Omega^2. \quad (3.6)$$

3.2.1 Killing vectors

A vector field ξ^μ is called a *Killing vector field* if the geometry is invariant under the infinitesimal transformations generated by ξ^μ . Mathematically, this condition is expressed by

$$\mathcal{L}_\xi g_{\mu\nu} = 0, \quad (3.7)$$

where $\mathcal{L}_\xi$ denotes the Lie derivative along ξ^μ . For the Levi–Civita connection this condition is equivalent to the Killing equation

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0. \quad (3.8)$$

If the spacetime admits a timelike Killing vector field, one can choose a time coordinate t adapted to its integral curves, so that

$$\xi = \partial_t. \quad (3.9)$$

In these adapted coordinates, the Killing condition implies

$$\partial_t g_{\mu\nu} = 0. \quad (3.10)$$

Thus the geometry is independent of the time coordinate. This property alone, however, defines a *stationary* spacetime and is not yet sufficient to guarantee that mixed time–space components of the metric vanish.

A static spacetime satisfies the stronger requirement that the timelike Killing vector field be *hypersurface-orthogonal*. Geometrically, this means that there exists a family of spacelike hypersurfaces everywhere orthogonal to the Killing flow. The corresponding integrability condition is the Frobenius condition

$$\xi_{[\mu} \nabla_\nu \xi_{\rho]} = 0. \quad (3.11)$$

When this condition is satisfied, one can locally introduce a scalar function t and a non-vanishing function N such that

$$\xi_\mu = -N \nabla_\mu t, \quad (3.12)$$

up to an irrelevant choice of sign and normalization. Since $\nabla_\mu t$ is normal to the hypersurfaces

$$t = \text{constant}, \quad (3.13)$$

Eq. (3.12) shows that the Killing vector itself is normal to these hypersurfaces.

To see explicitly why this eliminates the mixed time–space components, consider a vector X^μ tangent to a hypersurface $t = \text{constant}$. By definition,

$$X^\mu \nabla_\mu t = 0. \quad (3.14)$$

Using Eq. (3.12),

$$g_{\mu\nu} \xi^\mu X^\nu = \xi_\nu X^\nu = -N X^\nu \nabla_\nu t = 0. \quad (3.15)$$

Therefore the time-translation direction is orthogonal to every spatial direction tangent to the constant- t hypersurfaces.

Choosing the spatial coordinates inside these hypersurfaces and the time coordinate along the Killing flow, one has

$$\xi = \partial_t, \quad g(\partial_t, \partial_i) = 0, \quad (3.16)$$

and hence

$$\boxed{g_{ti} = 0.} \quad (3.17)$$

In the spherically symmetric metric considered here, spherical symmetry has already excluded mixed terms involving the angular coordinates, so the only remaining possible time–space cross term is $g_{tr} dt dr$. Staticity therefore allows us to choose coordinates for which

$$\boxed{g_{tr} = 0.} \quad (3.18)$$

The distinction between *stationary* and *static* is important. Stationarity requires only a timelike Killing vector and therefore only time-independence of the metric. Staticity additionally requires this Killing vector to be hypersurface-orthogonal. A rotating spacetime such as Kerr is stationary but not static: its time-translation Killing vector has non-vanishing twist and mixed time–angular terms cannot, in general, be removed globally.

In coordinates adapted to a static Killing field, the metric consequently has the form

$$ds^2 = g_{tt}(r) dt^2 + g_{rr}(r) dr^2 + r^2 d\Omega^2. \quad (3.19)$$

The familiar invariance under time reversal,

$$t \longrightarrow -t, \quad (3.20)$$

is now manifest. Since the coefficients of the metric are independent of t , while

$$dt \longrightarrow -dt, \quad (3.21)$$

the terms dt^2 , dr^2 and $d\Omega^2$ are unchanged. Therefore Eq. (3.19) is invariant¹ under $t \rightarrow -t$. It is better to regard this time-reversal symmetry as a consequence of the static form obtained from the hypersurface-orthogonal Killing field, rather than as the fundamental reason for imposing $g_{tr} = 0$.

We now use the signature convention $(-, +, +, +)$. In the static exterior region the Killing vector

$$\xi = \partial_t \quad (3.22)$$

is timelike. Therefore its norm must be negative,

$$g(\xi, \xi) < 0. \quad (3.23)$$

Since $\xi^\mu = (1, 0, 0, 0)$ in coordinates adapted to the Killing flow,

$$g(\xi, \xi) = g_{tt}, \quad (3.24)$$

¹By stating this we are assuming an eternal black, which is not consequence of a stellar collapse.

and hence

$$g_{tt}(r) < 0. \quad (3.25)$$

It is therefore convenient to define a positive function $A(r)$ by

$$A(r) \equiv -g_{tt}(r) > 0. \quad (3.26)$$

Similarly, the radial coordinate r is chosen inside the spacelike hypersurfaces $t = \text{constant}$. Thus ∂_r is spacelike in the static exterior region, and with the chosen signature

$$g(\partial_r, \partial_r) = g_{rr}(r) > 0. \quad (3.27)$$

We may therefore define

$$B(r) \equiv g_{rr}(r) > 0. \quad (3.28)$$

Hence the most general static and spherically symmetric metric in areal-radius coordinates can consequently be written as

$$\boxed{ds^2 = -A(r) dt^2 + B(r) dr^2 + r^2 d\Omega^2.} \quad (3.29)$$

Since the functions $A(r)$ and $B(r)$ are positive in this region, they may equivalently be parametrized as

$$A(r) = e^{2\alpha(r)}, \quad B(r) = e^{2\beta(r)}, \quad (3.30)$$

which gives

$$\boxed{ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2.} \quad (3.31)$$

This exponential parametrization introduces no additional physical assumption; it is only a convenient way of representing two positive functions.

In addition, at this stage $\alpha(r)$ and $\beta(r)$ are independent functions. In particular, no relation between g_{tt} and g_{rr} has yet been assumed. The relation

$$g_{tt}g_{rr} = -1 \quad (3.32)$$

will instead be derived later from the Einstein–Maxwell equations.

Deducing now both $A(r)$ and $B(r)$ we have to invoke the physical requirement of asymptotic flatness. This means that, far from the source, the geometry approaches that of Minkowski spacetime. In particular, the metric must approach the Minkowski form in spherical coordinates:

$$A(r) \longrightarrow 1, \quad B(r) \longrightarrow 1 \quad (r \rightarrow \infty), \quad (3.33)$$

and, after fixing the normalization of the time coordinate at infinity, this is equivalent to

$$\alpha(r) \longrightarrow 0, \quad \beta(r) \longrightarrow 0 \quad (r \rightarrow \infty). \quad (3.34)$$

3.2.2 Electromagnetic ansatz

The electromagnetic ansatz follows from the same symmetry argument. The most general static $SO(3)$ -invariant electromagnetic 2-form is

$$F = E(r) dt \wedge dr + P(r) \sin \theta d\theta \wedge d\phi. \quad (3.35)$$

The first term is the radial electric sector. The second is proportional to the invariant area 2-form on the symmetry spheres and represents a monopole-like magnetic sector. No mixed terms such as

$dt \wedge d\theta$ or $dr \wedge d\phi$ are allowed, because they would again define a preferred tangent direction on the sphere.

The homogeneous Maxwell equation $dF = 0$ further implies

$$P'(r) = 0, \quad (3.36)$$

so the magnetic coefficient is a constant magnetic charge P .

In the purely electric case considered by the simulator we set $P = 0$, and therefore

$$F = F_{tr}(r) dt \wedge dr. \quad (3.37)$$

Setting $P = 0$ is a physical choice of the model, not a consequence of spherical symmetry. If magnetic charge were retained, the final metric would depend on the combination $Q^2 + P^2$.

3.3 First step: solving Maxwell's equation

The determinant of the ansatz (3.31) is

$$g = -e^{2(\alpha+\beta)} r^4 \sin^2 \theta, \quad \sqrt{-g} = e^{\alpha+\beta} r^2 \sin \theta. \quad (3.38)$$

The homogeneous equation is automatically satisfied because F_{tr} depends only on r and $dF = 0$. The source-free inhomogeneous equation for $\nu = t$ instead becomes

$$0 = \nabla_\mu F^{t\mu} = \frac{1}{\sqrt{-g}} \partial_r (\sqrt{-g} F^{tr}).^2 \quad (3.39)$$

Multiplying by $\sqrt{-g}$ and integrating with respect to r gives

$$\sqrt{-g} F^{tr} = C(\theta), \quad (3.40)$$

where the integration constant is independent of r . Since

$$\sqrt{-g} = e^{\alpha+\beta} r^2 \sin \theta \quad (3.41)$$

and spherical symmetry requires F^{tr} to depend only on r , the entire angular dependence of the left-hand side is contained in $\sin \theta$. Therefore

$$C(\theta) = C_0 \sin \theta, \quad (3.42)$$

where C_0 is a constant.

The physical meaning of this integration constant follows from Gauss' law. In geometrized Gaussian units, the electric charge enclosed by a symmetry two-sphere is defined by the electric flux

$$Q \equiv \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sqrt{-g} F^{tr}. \quad (3.43)$$

Using

$$\sqrt{-g} F^{tr} = C_0 \sin \theta, \quad (3.44)$$

we obtain

$$Q = \frac{C_0}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \quad (3.45)$$

$$= \frac{C_0}{4\pi} (4\pi) = C_0. \quad (3.46)$$

²The idea behind this equation is the same of Eq (1.40)

Thus the integration constant is precisely the conserved electric charge, and we may write

$$\boxed{C(\theta) = Q \sin \theta.} \quad (3.47)$$

A non-zero Q is fully compatible with electrovacuum. The condition

$$J^\mu = 0 \quad (3.48)$$

means that there is no local electric four-current in the spacetime region under consideration. It does not imply that the electric flux through a closed sphere must vanish.

Indeed, in electrovacuum Maxwell's equation is

$$\nabla_\mu F^{\mu\nu} = 0. \quad (3.49)$$

For the static and spherically symmetric electric field considered here, the t component is equivalently

$$\partial_r \left(\sqrt{-g} F^{tr} \right) = 0. \quad (3.50)$$

Therefore the quantity $\sqrt{-g} F^{tr}$, and hence the electric flux through a symmetry sphere, is independent of r .

More explicitly, for two symmetry spheres of radii r_1 and r_2 contained in the same electrovacuum region,

$$\int_0^{2\pi} d\phi \int_0^\pi d\theta \sqrt{-g} F^{tr} \Big|_{r=r_1} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sqrt{-g} F^{tr} \Big|_{r=r_2}. \quad (3.51)$$

Thus Maxwell's equation requires the flux to be conserved, not to vanish.

In the Reissner–Nordström exterior, the central region is not part of the electrovacuum domain on which the source-free equations are being solved. Consequently, the absence of a local current,

$$J^\mu = 0, \quad (3.52)$$

is perfectly compatible with a non-zero conserved electric flux and hence with a non-zero total charge Q .

Finally,

$$F^{tr} = g^{tt} g^{rr} F_{tr} = -e^{-2(\alpha+\beta)} F_{tr}. \quad (3.53)$$

Using

$$\sqrt{-g} F^{tr} = Q \sin \theta, \quad (3.54)$$

together with

$$\sqrt{-g} = e^{\alpha+\beta} r^2 \sin \theta, \quad (3.55)$$

gives

$$\boxed{F_{tr} = -\frac{Q}{r^2} e^{\alpha+\beta}, \quad F^{tr} = \frac{Q}{r^2} e^{-(\alpha+\beta)}.} \quad (3.56)$$

Multiplying by $\sqrt{-g}$ and integrating with respect to r gives

$$\sqrt{-g} F^{tr} = C(\theta), \quad (3.57)$$

where the integration constant is independent of r . Since

$$\sqrt{-g} = e^{\alpha+\beta} r^2 \sin \theta \quad (3.58)$$

and spherical symmetry requires F^{tr} to depend only on r , the entire angular dependence of the left-hand side is contained in $\sin \theta$. Therefore

$$C(\theta) = C_0 \sin \theta, \quad (3.59)$$

where C_0 is a constant.

The physical meaning of this integration constant follows from Gauss' law. In geometrized Gaussian units the total electric charge measured through a symmetry two-sphere is

$$Q \equiv \frac{1}{4\pi} \int_{S^2} \star F. \quad (3.60)$$

With the orientation adopted here,

$$(\star F)_{\theta\phi} = \sqrt{-g} F^{tr}, \quad (3.61)$$

and hence

$$Q = \frac{1}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sqrt{-g} F^{tr} \quad (3.62)$$

$$= \frac{C_0}{4\pi} \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta = C_0. \quad (3.63)$$

Thus the integration constant is precisely the conserved electric charge, and we may write

$$\boxed{C(\theta) = Q \sin \theta.} \quad (3.64)$$

A non-zero Q is fully compatible with electrovacuum. The condition $J^\mu = 0$ states that there is no local electric current in the region under consideration; it does not require the electromagnetic flux through a sphere to vanish. Indeed, the source-free Maxwell equation

$$d \star F = 0 \quad (3.65)$$

implies that this flux is independent of the chosen sphere. It would be forced to vanish only if the sphere bounded a regular three-dimensional region entirely contained within the electrovacuum domain. In the Reissner–Nordström exterior the central region is excluded from that domain, so a non-zero conserved flux, and therefore a non-zero global charge Q , is allowed.

Moreover,

$$F^{tr} = g^{tt} g^{rr} F_{tr} = -e^{-2(\alpha+\beta)} F_{tr}. \quad (3.66)$$

Using $\sqrt{-g} F^{tr} = Q \sin \theta$ then gives

$$\boxed{F_{tr} = -\frac{Q}{r^2} e^{\alpha+\beta}, \quad F^{tr} = \frac{Q}{r^2} e^{-(\alpha+\beta)}.} \quad (3.67)$$

Multiplying by $\sqrt{-g}$ and integrating with respect to r gives

$$\sqrt{-g} F^{tr} = C(\theta). \quad (3.68)$$

The angular dependence of the determinant, together with spherical symmetry, requires $C(\theta) = Q \sin \theta$. Moreover,

$$F^{tr} = g^{tt} g^{rr} F_{tr} = -e^{-2(\alpha+\beta)} F_{tr}. \quad (3.69)$$

Substituting these two relations gives

$$\boxed{F_{tr} = -\frac{Q}{r^2} e^{\alpha+\beta}, \quad F^{tr} = \frac{Q}{r^2} e^{-(\alpha+\beta)}.} \quad (3.70)$$

The sign has been chosen so that, once $\alpha + \beta = 0$ has been obtained, the potential is $A = -Q dt/r$. Simultaneously reversing the signs of Q and F does not change the geometry.

The $1/r^2$ dependence was not imposed by hand. It follows from flux conservation: the flux through a sphere is constant while its area grows as r^2 . The metric factors $e^{\pm(\alpha+\beta)}$ account for the fact that covariant and contravariant components do not coincide in curved geometry.

3.4 Second step: constructing $T^\mu{}_\nu$

From (3.70), the quadratic invariant is

$$F_{\rho\sigma}F^{\rho\sigma} = 2F_{tr}F^{tr} = -\frac{2Q^2}{r^4}. \quad (3.71)$$

For example, the mixed time component is

$$\begin{aligned} T^t{}_t &= \frac{1}{4\pi} \left(F^{t\rho}F_{t\rho} - \frac{1}{4}F_{\rho\sigma}F^{\rho\sigma} \right) \\ &= \frac{1}{4\pi} \left(-\frac{Q^2}{r^4} + \frac{Q^2}{2r^4} \right) = -\frac{Q^2}{8\pi r^4}. \end{aligned} \quad (3.72)$$

The same calculation for the other indices yields

$$T^\mu{}_\nu = \frac{Q^2}{8\pi r^4} \text{diag}(-1, -1, +1, +1). \quad (3.73)$$

The sign structure has a physical meaning. The energy measured by a static observer is positive,

$$\rho_{\text{EM}} = -T^t{}_t = \frac{Q^2}{8\pi r^4}, \quad (3.74)$$

whereas the radial pressure is a tension,

$$p_r = T^r{}_r = -\rho_{\text{EM}}, \quad (3.75)$$

and the tangential pressures are positive,

$$p_\theta = p_\phi = +\rho_{\text{EM}}. \quad (3.76)$$

The sum $-\rho_{\text{EM}} + p_r + p_\theta + p_\phi$ vanishes, as required by the vanishing trace of the electromagnetic tensor.

3.5 Third step: Einstein's equations

Introduce

$$f(r) = e^{-2\beta(r)}. \quad (3.77)$$

This definition is equivalent to $g_{rr} = 1/f$. For the metric (3.31), the two useful independent components of the Einstein tensor are

$$G^t{}_t = \frac{f'}{r} + \frac{f-1}{r^2}, \quad (3.78)$$

$$G^r{}_r = \frac{2f\alpha'}{r} + \frac{f-1}{r^2}. \quad (3.79)$$

Using (3.73), the equations $G^\mu{}_\nu = 8\pi T^\mu{}_\nu$ therefore give

$$\frac{f'}{r} + \frac{f-1}{r^2} = -\frac{Q^2}{r^4}, \quad (3.80)$$

$$\frac{2f\alpha'}{r} + \frac{f-1}{r^2} = -\frac{Q^2}{r^4}. \quad (3.81)$$

3.5.1 Determination of $g_{tt}g_{rr}$

The electromagnetic tensor satisfies $T^t_t = T^r_r$. Consequently the two geometrical sides must also be equal. Subtracting (3.81) from (3.80) gives

$$f' - 2f\alpha' = 0. \quad (3.82)$$

In regions where $f \neq 0$, division by f gives

$$\frac{f'}{f} = 2\alpha'. \quad (3.83)$$

Integrating,

$$2\alpha = \ln f + \text{constant}. \quad (3.84)$$

The constant would multiply dt^2 by a constant factor and is removed by rescaling t ; asymptotic flatness fixes this rescaling by requiring $e^{2\alpha} \rightarrow 1$. Therefore

$$e^{2\alpha} = f = e^{-2\beta}, \quad g_{tt}g_{rr} = -1. \quad (3.85)$$

3.5.2 Integration of the radial equation

Multiplying (3.80) by r^2 gives

$$rf' + f - 1 = -\frac{Q^2}{r^2}. \quad (3.86)$$

Since $(rf)' = rf' + f$,

$$(rf)' = 1 - \frac{Q^2}{r^2}. \quad (3.87)$$

Integrating,

$$rf = r + \frac{Q^2}{r} + C, \quad f = 1 + \frac{C}{r} + \frac{Q^2}{r^2}. \quad (3.88)$$

Integration has introduced a constant C . To identify it, require the time component at large distance to contain the ADM mass:

$$g_{tt} = -f = -1 + \frac{2M}{r} + \mathcal{O}(r^{-2}). \quad (3.89)$$

This fixes $C = -2M$. Hence

$$\boxed{f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}}. \quad (3.90)$$

The angular component supplies an independent check. After imposing $e^{2\alpha} = f$,

$$G^\theta_\theta = G^\phi_\phi = \frac{1}{2}f'' + \frac{f'}{r}. \quad (3.91)$$

Substituting (3.90),

$$G^\theta_\theta = \frac{Q^2}{r^4} = 8\pi T^\theta_\theta. \quad (3.92)$$

The solution therefore also satisfies the equation not used in the integration; in general, this compatibility is guaranteed by the Bianchi identity together with Maxwell's equations.

3.5.3 Interpretation through the mass function

It is useful to rewrite

$$f(r) = 1 - \frac{2m(r)}{r}. \quad (3.93)$$

Equation (3.80) is then equivalent to

$$m'(r) = 4\pi r^2 \rho_{\text{EM}} = \frac{Q^2}{2r^2}. \quad (3.94)$$

Integrating towards infinity and imposing $m(\infty) = M$ gives

$$m(r) = M - \frac{Q^2}{2r}. \quad (3.95)$$

The Q^2/r^2 term in the metric thus expresses the energy of the electromagnetic field distributed outside radius r . This interpretation does not replace the preceding derivation, but clarifies its energetic content.

3.6 Final solution and potential

The Reissner–Nordström metric is

$$\boxed{ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}.} \quad (3.96)$$

A gauge choice for the four-potential is

$$A = -\frac{Q}{r}dt, \quad F = dA, \quad (3.97)$$

and it produces precisely $F_{tr} = -Q/r^2$ after using $\alpha + \beta = 0$. An additive constant in A_t is a gauge transformation and is normally chosen so that $A_t \rightarrow 0$ at infinity.

Restoring SI units, with $t_g = ct_{\text{SI}}$,

$$f(r) = 1 - \frac{2GM_{\text{SI}}}{c^2 r} + \frac{GQ_{\text{SI}}^2}{4\pi\epsilon_0 c^4 r^2}. \quad (3.98)$$

For intuition

The term $+Q^2/r^2$ should not be interpreted as a “repulsive electric force” acting on every particle. A neutral particle also follows geodesics of the modified metric. Repulsion or attraction of a *charged* particle would additionally require the Lorentz term $qF^\mu{}_\nu u^\nu$, which the Web Demo does not integrate.

3.6.1 Three immediate checks

The solution must pass at least three checks:

1. For $Q = 0$, Schwarzschild is recovered, $f = 1 - 2M/r$;
2. For $M = Q = 0$, Minkowski spacetime in spherical coordinates is recovered;
3. For $r \rightarrow \infty$, the leading correction is $2M/r$, while the field approaches the Coulomb field with total charge Q .

These limits identify the constants of the solution and prevent a convention-dependent artefact from being mistaken for a physical parameter.

3.7 Coordinate singularities and curvature singularities

The vanishing trace of $T_{\mu\nu}^{\text{EM}}$ implies $R = 0$, but spacetime is not flat. The Kretschmann invariant is

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48M^2}{r^6} - \frac{96MQ^2}{r^7} + \frac{56Q^4}{r^8}. \quad (3.99)$$

It diverges only at $r = 0$. By contrast, the divergences of g_{rr} at points where $f = 0$ can be removed with regular coordinates: they signal horizons, not curvature singularities [1, 5].

The two Reissner–Nordström horizons

4.1 The geometrical question

The metric (3.96) is expressed in a static chart that becomes problematic when $f(r) = 0$. Before calling these zeros “horizons”, three distinct facts must be separated:

1. A metric coefficient may diverge because the coordinate chart fails;
2. A hypersurface is null if its normal has zero norm;
3. An event horizon is defined globally by the causal structure, not by the isolated vanishing of a metric component.

This order will be followed. First the candidate surfaces will be found, then their null character will be verified, regular coordinates will be constructed, and only then will r_+ and r_- be interpreted.

4.2 Roots and factorization

Since $r > 0$ in the ordinary areal region, the zeros of f coincide with those of the numerator

$$r^2 f(r) = r^2 - 2Mr + Q^2. \quad (4.1)$$

The quadratic formula gives

$$r_{\pm} = M \pm \sqrt{M^2 - Q^2}. \quad (4.2)$$

The roots are real if and only if $M^2 \geq Q^2$. In the subextremal case $M^2 > Q^2$, they are distinct and allow the factorization

$$f(r) = \frac{(r - r_+)(r - r_-)}{r^2}, \quad r_+ + r_- = 2M, \quad r_+ r_- = Q^2. \quad (4.3)$$

Since the denominator is positive for $r > 0$, the sign of f is determined by the two linear factors. Hence

Region	$r > r_+$	$r_- < r < r_+$	$0 < r < r_-$
$f(r)$	+	−	+
t	timelike	spacelike	timelike
r	spacelike	timelike	spacelike

4.3 The outer horizon r_+

In the exterior region $r > r_+$ one has $f(r) > 0$. With the signature $(-, +, +, +)$,

$$g_{tt} = -f < 0, \quad g_{rr} = \frac{1}{f} > 0, \quad (4.4)$$

so that ∂_t is timelike and ∂_r is spacelike. Thus t plays the role of a time coordinate while r is a spatial coordinate.

Between the two roots,

$$r_- < r < r_+, \quad (4.5)$$

one instead has $f(r) < 0$, and therefore

$$g_{tt} > 0, \quad g_{rr} < 0. \quad (4.6)$$

Hence ∂_t becomes spacelike whereas ∂_r becomes timelike. The coordinate r therefore plays a temporal role in this region.

The sign change of f , however, determines only the causal character of the coordinate directions. It does not by itself determine whether increasing or decreasing r corresponds to the future direction. A time orientation must also be specified. We will consider the black-hole sector reached by crossing the *future* outer horizon from the chosen asymptotically flat exterior. After introducing coordinates regular across $r = r_+$, we will show explicitly that future-directed causal curves in this sector evolve towards decreasing values of r .

It is also important that the vanishing of $f(r)$ alone does not define an event horizon. Relative to the chosen asymptotically flat exterior, the future event horizon is defined globally as the boundary of the causal past of future null infinity,

$$\boxed{\mathcal{H}^+ = \partial J^-(\mathcal{I}^+).} \quad (4.7)$$

The surface $r = r_+$ coincides with this boundary in the maximally extended Reissner–Nordström geometry. The local description in terms of the sign of f is consistent with this global definition, but does not replace it.

The limitations of the static coordinates can already be seen by considering a radial null ray. Setting $d\Omega = 0$ and $ds^2 = 0$ gives

$$0 = -f dt^2 + f^{-1} dr^2 \implies \frac{dr}{dt} = \pm f. \quad (4.8)$$

Since $f \rightarrow 0$ as $r \rightarrow r_+$, the coordinate velocity dr/dt tends to zero and the Schwarzschild-like coordinate time t required to reach the horizon diverges. This does not describe a physical halt of the infalling observer: its proper time remains finite. The divergence instead signals the breakdown of the static time coordinate at the horizon.

4.4 Tortoise coordinate and extension across r_+

Define the tortoise coordinate r_* by

$$\frac{dr_*}{dr} = \frac{1}{f(r)}. \quad (4.9)$$

For $r_+ \neq r_-$, the partial-fraction decomposition is

$$\frac{1}{f(r)} = \frac{r^2}{(r - r_+)(r - r_-)} = 1 + \frac{r_+^2}{r_+ - r_-} \frac{1}{r - r_+} - \frac{r_-^2}{r_+ - r_-} \frac{1}{r - r_-}. \quad (4.10)$$

Integration term by term gives

$$r_* = r + \frac{r_+^2}{r_+ - r_-} \ln \left| \frac{r}{r_+} - 1 \right| - \frac{r_-^2}{r_+ - r_-} \ln \left| \frac{r}{r_-} - 1 \right| + C. \quad (4.11)$$

Now introduce the advanced Eddington–Finkelstein coordinate

$$v = t + r_*, \quad dt = dv - \frac{dr}{f}. \quad (4.12)$$

Substitution into the t – r part of the metric gives

$$\begin{aligned} -f dt^2 + f^{-1} dr^2 &= -f \left(dv - \frac{dr}{f} \right)^2 + f^{-1} dr^2 \\ &= -f dv^2 + 2 dv dr. \end{aligned} \quad (4.13)$$

The metric therefore becomes

$$\boxed{ds^2 = -f(r) dv^2 + 2 dv dr + r^2 d\Omega^2.} \quad (4.14)$$

All metric coefficients remain finite at $f(r) = 0$. Thus the divergence of the static components at $r = r_+$ is a coordinate singularity rather than a curvature singularity.

4.4.1 Future direction inside the outer horizon

The regular Eddington–Finkelstein chart also makes the causal orientation inside the outer horizon explicit. Let $x^\mu(\lambda)$ be a future-directed causal curve, with tangent vector $\dot{x}^\mu$. From Eq. (4.14), the causal condition

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \leq 0 \quad (4.15)$$

becomes

$$-f \dot{v}^2 + 2 \dot{v} \dot{r} + r^2 \left(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right) \leq 0. \quad (4.16)$$

We choose the future orientation inherited continuously from the original asymptotically flat exterior. For a generic future-directed causal curve, $\dot{v} > 0$. In the black-hole region

$$r_- < r < r_+, \quad f < 0, \quad (4.17)$$

Eq. (4.16) gives

$$2 \dot{v} \dot{r} \leq f \dot{v}^2 - r^2 \left(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2 \right) < 0. \quad (4.18)$$

Since $\dot{v} > 0$, one obtains

$$\boxed{\dot{r} < 0.} \quad (4.19)$$

The limiting case $\dot{v} = 0$ corresponds to radial null curves with $v = \text{constant}$. Their future-directed branch is likewise the ingoing branch, for which $\dot{r} < 0$. Therefore every future-directed causal curve in the black-hole sector evolves towards decreasing values of the areal radius.

This gives a precise meaning to the statement that r is timelike between the two horizons. In this region, reversing from decreasing to increasing r is not analogous to reversing an ordinary spatial velocity: it would amount to changing the local time orientation.

This statement must be distinguished from the white-hole sector of the maximal eternal extension. There too

$$r_- < r < r_+, \quad f < 0, \quad (4.20)$$

and r is timelike, but the future orientation corresponds to increasing rather than decreasing r . Thus the sign of f determines the causal character of the coordinate r , whereas the choice of black-hole or white-hole sector determines which direction along r is future-directed.

Consequently, once a future-directed causal curve crosses $r = r_+$ from the chosen exterior into the black-hole sector, it cannot return to the same exterior while remaining future-directed.

4.4.2 Kruskal-type coordinates and surface gravity

Kruskal-type coordinates can be constructed by exponentiating suitable null coordinates with a scale fixed by the surface gravity of the corresponding horizon. For a simple root r_h of f ,

$$\kappa_h = \frac{1}{2}f'(r_h). \quad (4.21)$$

Using

$$r_+ r_- = Q^2, \quad r_+ + r_- = 2M, \quad (4.22)$$

one obtains

$$\kappa_+ = \frac{r_+ - r_-}{2r_+^2}, \quad \kappa_- = -\frac{r_+ - r_-}{2r_-^2}. \quad (4.23)$$

The magnitudes of the two surface gravities are different. A single exponentiation adapted to one horizon therefore does not simultaneously regularize the other; separate Kruskal-type charts adapted to r_+ and r_- are required.

4.5 The inner horizon r_- : the Cauchy horizon

After crossing r_- , the sign of f becomes positive again:

$$0 < r < r_- \quad \implies \quad f(r) > 0. \quad (4.24)$$

Consequently, ∂_r becomes spacelike once more while ∂_t becomes timelike. Thus the causal character of the coordinates returns to the same type as in the exterior region: changes in r again describe spatial motion rather than an unavoidable evolution towards the future.

This is fundamentally different from the region $r_- < r < r_+$, where r is timelike and every future-directed causal curve in the black-hole sector is forced towards decreasing r . Once the inner horizon has been crossed, future-directed motion is no longer constrained to proceed towards smaller values of the areal radius. A causal observer may therefore move through the region $0 < r < r_-$ without necessarily reaching $r = 0$.

The curvature singularity at $r = 0$ is correspondingly timelike, rather than spacelike as in Schwarzschild. It should therefore be thought of as a singular boundary located in a spatial direction of this inner region, not as a singular surface lying inevitably in the future of every infalling observer. In the eternal, exactly spherical and unperturbed Reissner–Nordström solution, suitable causal curves can avoid this timelike singularity and cross another branch of the inner horizon, continuing into further regions of the maximal analytic extension. The corresponding conformal diagram therefore contains a chain of asymptotically flat regions connected through black-hole and white-hole sectors [1].

To understand the name “Cauchy horizon”, consider initial data specified on a spacelike partial Cauchy hypersurface that extends through the chosen exterior and into the black-hole region but does not cross the inner horizon. The hyperbolic field equations determine the evolution uniquely only inside the domain of dependence of this hypersurface. The future boundary of that domain is the inner horizon $r = r_-$. Beyond this surface, causal influences can enter from regions of the maximal extension that were not represented in the original initial data.

Therefore:

- r_+ is the outer future event horizon relative to the chosen asymptotically flat exterior;
- r_- is an inner null horizon and, in the maximal eternal extension, a *Cauchy horizon*: beyond it, the data specified on the chosen partial Cauchy hypersurface are no longer sufficient to determine the evolution uniquely.

The eternal extension is not a realistic model of the interior

The smooth continuation through r_- described above is a property of the exact, eternal Reissner–Nordström geometry. In a realistic black hole, however, even arbitrarily small perturbations are strongly blueshifted as they approach the inner horizon. Ingoing and outgoing fluxes interact, causing an effective internal mass parameter to grow without bound in the classical analysis, a phenomenon known as *mass inflation*. The Cauchy horizon of the exact Reissner–Nordström solution is therefore unstable under generic perturbations [10]. The maximally extended diagram remains essential for understanding the exact solution, but its smooth inner continuation should not be identified directly with the interior of a black hole formed by realistic gravitational collapse.

4.6 The three regimes

4.6.1 Subextremal: $M^2 > Q^2$

There are two distinct real roots. The function f changes sign at each root and the surface gravities are nonzero. This is the case with outer and inner horizons discussed above.

4.6.2 Extremal: $M^2 = Q^2$

The roots coincide:

$$r_+ = r_- = M, \quad f(r) = \left(1 - \frac{M}{r}\right)^2. \quad (4.25)$$

The sign of f does not change across $r = M$; r remains spacelike on both sides. The root is double, the surface gravity vanishes, and the tortoise coordinate contains a pole:

$$r_* = r + 2M \ln |r - M| - \frac{M^2}{r - M} + C. \quad (4.26)$$

The geometry develops an infinitely long proper throat in the static spatial section. The extremal limit is not obtained merely by ignoring the distinction between the two horizons: the global structure changes qualitatively.

4.6.3 Superextremal: $M^2 < Q^2$

There are no real roots of f . The timelike curvature singularity at $r = 0$ is not covered by a horizon: the solution contains a naked singularity. The simulator flags this regime and disables the horizon display.

Why two roots do not mean two identical barriers

r_+ and r_- are both null hypersurfaces and Killing horizons of the exact solution, but their global roles differ. r_+ separates the causal exterior from the black hole; r_- bounds the domain determined by exterior data and is unstable to blueshift. Calling them simply the “large horizon” and “small horizon” would conceal the fundamental physical difference.

The solver uses Boyer–Lindquist coordinates, which in the absence of rotation coincide with the static Reissner–Nordström coordinates. In this chart g_{rr} diverges at r_+ . To avoid numerical instability, integration is stopped at $r \leq r_+ + 0.12M$. This stopping rule is a numerical choice tied to the chart, not a physical barrier for a falling observer.

4.7 Photon orbit in Reissner–Nordström

For an equatorial null geodesic, the effective potential is proportional to

$$V_{\text{eff}}(r) = L_z^2 \frac{f(r)}{r^2}. \quad (4.27)$$

A circular orbit requires $d(f/r^2)/dr = 0$, namely

$$r^2 - 3Mr + 2Q^2 = 0. \quad (4.28)$$

The two formal roots are

$$r_{\gamma\pm} = \frac{3M \pm \sqrt{9M^2 - 8Q^2}}{2}. \quad (4.29)$$

The Web Demo displays the outer root

$$r_\gamma = \frac{3M + \sqrt{9M^2 - 8Q^2}}{2}, \quad (4.30)$$

which yields $r_\gamma = 3M$ for $Q = 0$.

4.8 Physical interpretation: collapse and eternal solutions

A final conceptual distinction is important. The Schwarzschild and Reissner–Nordström metrics may be interpreted either as exterior geometries generated by a material source or as parts of maximally extended eternal spacetimes. These two interpretations have the same local metric in the appropriate region, but they correspond to different global spacetimes.

For a black hole formed by gravitational collapse, the parameters M and Q have a direct physical history. The mass and, in the charged case, the electric charge are carried by the collapsing matter. After this matter crosses the event horizon, the exterior region may be vacuum in the Schwarzschild case,

$$T_{\mu\nu} = 0, \quad (4.31)$$

or electrovacuum in the Reissner–Nordström case,

$$J^\mu = 0, \quad T_{\mu\nu}^{\text{EM}} \neq 0. \quad (4.32)$$

Nevertheless, an exterior observer still measures the total mass M and, when present, the total charge Q . Thus the absence of matter or electric current in the exterior does not imply that these global quantities vanish.

The maximally extended eternal solutions have a different interpretation. They are not obtained by describing the dynamical collapse of a material body, but by analytically extending the exact vacuum or electrovacuum solution through every removable coordinate singularity. Consequently, in eternal Schwarzschild there is no regular material source carrying the mass M , and in eternal

Reissner–Nordström there is no regular electric current carrying the charge Q . Instead, M and Q are global parameters of the solution: M is identified through the asymptotic gravitational field, while Q is measured through the electric flux at infinity.

Therefore, a black hole produced by collapse must not be identified globally with the corresponding maximally extended eternal geometry. In particular, the additional white-hole regions and asymptotic regions of the eternal extension are not generated by ordinary gravitational collapse. For Reissner–Nordström this distinction is even more important because the inner Cauchy horizon of the exact eternal solution is unstable under generic perturbations, so the ideal maximal extension should not be interpreted literally as the interior structure of a realistic charged black hole.

From spherical symmetry to rotation

The aim of this chapter is to understand how the static, spherically symmetric black-hole geometries discussed so far are generalized when angular momentum is introduced. Starting from Schwarzschild and Reissner–Nordström, we move towards Kerr and Kerr–Newman by identifying which symmetries survive rotation, which ones are lost, and how the angular momentum parameter enters the geometry.

The final goal is not merely to rewrite a static metric in rotating coordinates, but to describe a genuinely rotating spacetime carrying a non-zero conserved angular momentum J . In the charged case, the resulting geometry will be the Kerr–Newman solution, characterized asymptotically by the three global parameters

$$M, \quad Q, \quad J, \quad (5.1)$$

5.1 Staticity, stationarity, and axisymmetry

A geometry is stationary if a timelike Killing field ξ exists in the asymptotic region such that

$$\mathcal{L}_\xi g_{\mu\nu} = 0. \quad (5.2)$$

It is static if, in addition, ξ is orthogonal to a family of hypersurfaces. By the Frobenius theorem, this second condition can be expressed as

$$\xi_{[\mu} \nabla_\nu \xi_{\rho]} = 0. \quad (5.3)$$

Schwarzschild and Reissner–Nordström satisfy both conditions and admit the full three-dimensional rotation group. Introducing genuine angular momentum breaks spherical symmetry while preserving invariance under rotations about a single preferred axis. Kerr and Kerr–Newman therefore retain stationarity and axisymmetry, but are no longer static.

There are two manifest Killing fields,

$$\xi = \partial_t, \quad \eta = \partial_\phi, \quad (5.4)$$

associated respectively with time translations and rotations about the symmetry axis; the orbits of η are closed with period 2π . The metric coefficients depend on neither t nor ϕ , but $g_{t\phi} \neq 0$, and the Frobenius condition for ξ fails when $a \neq 0$: spacetime is stationary but not static.

For intuition

“Stationary” means that the geometry does not change in time; “static” additionally requires the absence of gravitomagnetic circulation. A steady water vortex is an imperfect but useful analogy: its profile does not change, yet the fluid rotates.

5.2 Why rotating the coordinates is not enough

Before constructing the rotating solution, it is important to distinguish between a genuinely rotating geometry and a static geometry described in a rotating coordinate system.

Applying a transformation

$$\phi' = \phi - \Omega t \quad (5.5)$$

to the Schwarzschild metric produces mixed terms $dt d\phi'$. The same happens even in flat spacetime described by rotating observers. The appearance of a mixed term is therefore not, by itself, evidence of physical angular momentum: a coordinate transformation changes neither the curvature tensor nor the asymptotic charges.

In a stationary and axisymmetric geometry, however, the mixed component $g_{t\phi}$ has a direct geometrical interpretation once the coordinates are normalized with respect to non-rotating observers at spatial infinity. Since the line element contains

$$2g_{t\phi} dt d\phi, \quad (5.6)$$

a non-zero $g_{t\phi}$ expresses a coupling between the temporal Killing direction ∂_t and the axial Killing direction ∂_ϕ . This coupling is responsible for the gravitomagnetic effect usually called *frame dragging*. Nevertheless, $g_{t\phi} \neq 0$ by itself is not coordinate invariant, as the rotating-coordinate example above already shows.

The physical angular momentum is instead a global charge associated with the axial Killing field

$$\eta = \partial_\phi. \quad (5.7)$$

For an asymptotically flat stationary and axisymmetric spacetime, its total value can be defined at spatial infinity through the Komar integral

$$J = \frac{1}{16\pi} \int_{S_\infty^2} \nabla^\mu \eta^\nu dS_{\mu\nu}, \quad (5.8)$$

in geometrized units and with the chosen orientation convention. Here $dS_{\mu\nu}$ is the antisymmetric surface element of the two-sphere at spatial infinity. Thus J is not introduced through a particular metric component: it is the conserved geometrical charge associated with rotational symmetry.

The same quantity can be read directly from the asymptotic metric. In asymptotically non-rotating coordinates,

$$g_{t\phi} = -\frac{2J}{r} \sin^2 \theta + \mathcal{O}(r^{-2}). \quad (5.9)$$

The coefficient of the leading $1/r$ rotational term therefore identifies the same angular momentum J measured by the Komar integral. In the Kerr and Kerr–Newman families one writes

$$J = Ma, \quad (5.10)$$

so that a measures the angular momentum per unit mass in geometrized units.

This distinction sharply separates a *rotating geometry* from a *rotating chart* on the same static geometry. A coordinate transformation may generate a non-zero mixed component, but it cannot generate a non-zero asymptotic angular momentum J .

The rotating solution must therefore be obtained as a genuinely different solution of Einstein's equations under the weaker assumptions of stationarity and axisymmetry. Kerr found the vacuum solution in 1963; Newman and collaborators obtained the charged generalization in 1965 [6, 7].

5.3 One route to the solution: Kerr–Schild and Newman–Janis

There is no direct analogue of the simple spherical derivation used for Schwarzschild and Reissner–Nordström. Rotation removes spherical symmetry and introduces a non-trivial coupling between the temporal and azimuthal directions. The purpose of this section is therefore not yet to introduce the coordinate system in which the final Kerr–Newman metric will be analysed. That geometrical interpretation, together with Boyer–Lindquist coordinates, will be developed in the following chapter. Here we only introduce a useful structural form from which the rotating solution can be constructed.

A particularly useful choice is the Kerr–Schild ansatz

$$\boxed{g_{\mu\nu} = \eta_{\mu\nu} + 2Hk_\mu k_\nu}, \quad \eta^{\mu\nu}k_\mu k_\nu = 0, \quad (5.11)$$

where $\eta_{\mu\nu}$ is a flat background metric, H is a scalar function, and k_μ is a null covector with respect to the flat background.

This ansatz is much more restrictive than a generic decomposition $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. The deformation has rank one and is built entirely from a null direction. If

$$k^\mu \equiv \eta^{\mu\nu}k_\nu, \quad (5.12)$$

then the null condition gives

$$\eta_{\mu\nu}k^\mu k^\nu = 0. \quad (5.13)$$

Moreover,

$$g_{\mu\nu}k^\mu k^\nu = \eta_{\mu\nu}k^\mu k^\nu + 2H(k_\mu k^\mu)^2 = 0, \quad (5.14)$$

so the same vector is null with respect to the full metric as well.

The null structure also makes the inverse metric unusually simple:

$$\boxed{g^{\mu\nu} = \eta^{\mu\nu} - 2Hk^\mu k^\nu}. \quad (5.15)$$

Indeed,

$$\begin{aligned} g_{\mu\rho}g^{\rho\nu} &= (\eta_{\mu\rho} + 2Hk_\mu k_\rho)(\eta^{\rho\nu} - 2Hk^\rho k^\nu) \\ &= \delta_\mu^\nu - 4H^2k_\mu(k_\rho k^\rho)k^\nu \\ &= \delta_\mu^\nu. \end{aligned} \quad (5.16)$$

This special algebraic structure is one of the reasons why the Kerr–Schild ansatz is so useful: although the full geometry is nonlinear, the dependence of the inverse metric and of the curvature on the deformation is considerably simpler than for a generic metric perturbation.

For Kerr–Newman, the scalar function takes the form

$$H = \frac{Mr - \frac{1}{2}Q^2}{\Sigma}, \quad \Sigma = r^2 + a^2 \cos^2 \theta. \quad (5.17)$$

Therefore

$$2H = \frac{2Mr - Q^2}{\Sigma}. \quad (5.18)$$

The same combination $2Mr - Q^2$ already encountered in the Reissner–Nordström geometry survives in the rotating solution, while the replacement

$$r^2 \longrightarrow \Sigma = r^2 + a^2 \cos^2 \theta \quad (5.19)$$

introduces the angular dependence generated by rotation.

The precise geometrical meaning of the coordinates appearing in this representation, and their relation to the Boyer–Lindquist coordinates used for the final form of the metric, will be discussed in the following chapter. At this stage, the Kerr–Schild form should be regarded primarily as a convenient structural representation of the rotating geometry.

A related historical route is provided by the Newman–Janis algorithm. Its role here is only to motivate how the Reissner–Nordström solution can be connected to Kerr–Newman; the coordinate systems and the resulting real geometry will be analysed explicitly afterwards. The procedure can be summarized schematically as follows:

1. Rewrite the static solution in an advanced null chart;
2. Express the inverse metric through a null tetrad;
3. Formally complexify the radial and null coordinates;
4. Introduce a complex shift containing $ia \cos \theta$;
5. Impose a real section and reconstruct a real metric.

The procedure produces the Kerr–Newman metric, but the resulting geometry must still be verified directly in the Einstein–Maxwell equations. One of the characteristic combinations generated by the construction is

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad (5.20)$$

which can formally be associated with

$$(r + ia \cos \theta)(r - ia \cos \theta) = r^2 + a^2 \cos^2 \theta. \quad (5.21)$$

What is and is not a derivation

The Newman–Janis procedure generates the correct solution, but the complexification is not an ordinary real diffeomorphism and is ambiguous when applied to arbitrary metrics. It should therefore not be interpreted as a coordinate transformation that turns Reissner–Nordström into Kerr–Newman.

For this reason, the following chapter separates the coordinate geometry from the existence of the solution itself: Boyer–Lindquist coordinates and their geometrical meaning will first be introduced explicitly, after which the already verified Kerr–Newman geometry can be written and analysed in that chart. For a detailed critical discussion of the Newman–Janis construction, see Adamo–Newman [11].

Boyer–Lindquist coordinates

The previous chapter explained why a rotating black hole cannot be obtained by merely introducing rotating coordinates on a static geometry. Kerr and Kerr–Newman are genuinely different stationary and axisymmetric solutions, characterized by a non-zero angular momentum $J = Ma$. We also introduced the Kerr–Schild structure and the Newman–Janis construction as routes leading to the rotating solution.

At this point we assume that the Kerr–Newman¹ geometry has already been obtained and verified as a solution of the Einstein–Maxwell equations. The purpose of the present chapter is different: we must choose coordinates in which its stationary and axial symmetries, its radial structure, and the constants of motion are as transparent as possible.

A horizon-penetrating advanced chart is particularly natural during the construction of the solution, but it contains mixed radial terms. The Boyer–Lindquist coordinates arise precisely from the attempt to remove these radial cross terms while preserving the coordinates adapted to the two Killing symmetries. They are therefore not guessed independently of the metric: they are constructed from the already known rotating geometry.

6.1 From advanced coordinates to Boyer–Lindquist coordinates

Define

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2. \quad (6.1)$$

Given the Kerr–Newman solution, it can be written in an advanced, horizon-penetrating chart

$$(v, r, \theta, \tilde{\phi}) \quad (6.2)$$

as

$$\begin{aligned} ds^2 = & - \left(1 - \frac{2Mr - Q^2}{\Sigma} \right) dv^2 + 2dvdr + \Sigma d\theta^2 - 2a \sin^2 \theta dr d\tilde{\phi} \\ & - \frac{2a(2Mr - Q^2) \sin^2 \theta}{\Sigma} dv d\tilde{\phi} \\ & + \left(r^2 + a^2 + \frac{a^2(2Mr - Q^2) \sin^2 \theta}{\Sigma} \right) \sin^2 \theta d\tilde{\phi}^2. \end{aligned} \quad (6.3)$$

Equation (6.3) is a coordinate representation of the Kerr–Newman solution motivated in the preceding chapter by the Kerr–Schild/Newman–Janis construction. The role of the present chapter is only to change its coordinate representation.

The advanced chart has an important advantage: it is regular across the future horizon. However, it contains the two radial mixed terms

$$dv dr, \quad dr d\tilde{\phi}. \quad (6.4)$$

For the study of stationary orbits and geodesics outside the black hole, it is more convenient to use a chart in which the radial direction is decoupled from the two symmetry directions.

We therefore seek new coordinates (t, r, θ, ϕ) satisfying three requirements:

¹This choice, although difficult, is based on the attempt to keep this document within 100 pages.

1. r and θ are left unchanged;
2. t and ϕ remain adapted to stationarity and axisymmetry;
3. the mixed terms $dt dr$ and $dr d\phi$ vanish.

A natural transformation compatible with these requirements is

$$dv = dt + \lambda(r)dr, \quad d\tilde{\phi} = d\phi + \chi(r)dr. \quad (6.5)$$

The restriction of λ and χ to functions of r alone is important: it preserves the independence of the metric coefficients from t and ϕ and therefore keeps the Killing fields manifest.

Let $g_{ab}^{(\text{adv})}$ denote the components of (6.3). After substitution, the two new mixed coefficients are

$$g_{tr}^{(\text{new})} = g_{vv}\lambda + g_{vr} + g_{v\tilde{\phi}}\chi, \quad (6.6)$$

$$g_{r\phi}^{(\text{new})} = g_{v\tilde{\phi}}\lambda + g_{r\tilde{\phi}} + g_{\tilde{\phi}\tilde{\phi}}\chi. \quad (6.7)$$

Requiring

$$g_{tr}^{(\text{new})} = 0, \quad g_{r\phi}^{(\text{new})} = 0 \quad (6.8)$$

gives the linear system

$$\begin{pmatrix} g_{vv} & g_{v\tilde{\phi}} \\ g_{v\tilde{\phi}} & g_{\tilde{\phi}\tilde{\phi}} \end{pmatrix} \begin{pmatrix} \lambda \\ \chi \end{pmatrix} = - \begin{pmatrix} g_{vr} \\ g_{r\tilde{\phi}} \end{pmatrix}. \quad (6.9)$$

Substituting the components of the advanced metric and using

$$r^2 + a^2 - (2Mr - Q^2) = \Delta, \quad (6.10)$$

the solution simplifies nontrivially to

$$\boxed{\lambda(r) = \frac{r^2 + a^2}{\Delta}, \quad \chi(r) = \frac{a}{\Delta}.} \quad (6.11)$$

Thus

$$\boxed{dv = dt + \frac{r^2 + a^2}{\Delta}dr, \quad d\tilde{\phi} = d\phi + \frac{a}{\Delta}dr.} \quad (6.12)$$

Because both coefficients depend only on r , the transformation is locally integrable:

$$v = t + \int \frac{r^2 + a^2}{\Delta} dr, \quad \tilde{\phi} = \phi + \int \frac{a}{\Delta} dr. \quad (6.13)$$

These equations define the Boyer–Lindquist time and azimuthal coordinates relative to the advanced chart.

Substituting (6.12) into (6.3), Eqs. (6.6)–(6.7) ensure the cancellation of the radial mixed terms. The remaining radial coefficient becomes

$$g_{rr} = \frac{\Sigma}{\Delta}, \quad (6.14)$$

and the only surviving off-diagonal component is $g_{t\phi}$, coupling the two Killing directions.

This is the defining practical feature of Boyer–Lindquist coordinates: they are coordinates adapted to stationarity and axisymmetry in which the radial and polar directions are diagonal, while the genuine rotational coupling between t and ϕ remains [8, 11].

The complete metric obtained by this transformation will be written explicitly and analysed in the following chapter.

The transformation also explains the singularity of the chart

The functions

$$\lambda = \frac{r^2 + a^2}{\Delta}, \quad \chi = \frac{a}{\Delta} \quad (6.15)$$

diverge when $\Delta = 0$. The starting advanced chart is regular on the future horizon; it is the attempt to eliminate the mixed radial terms that makes the Boyer–Lindquist chart singular there.

The divergence of

$$g_{rr} = \frac{\Sigma}{\Delta} \quad (6.16)$$

is therefore the price paid for the almost diagonal form of the metric, not a physical singularity of the geometry.

6.2 Geometrical meaning: oblate spheroidal coordinates

Having constructed the Boyer–Lindquist chart from the rotating geometry, we can now understand the geometrical meaning of its coordinates. In particular, the coordinate r should not be interpreted as the ordinary spherical radius.

The distinction can already be seen in the flat limit. Set

$$M = Q = 0 \quad (6.17)$$

while retaining a as a coordinate parameter. The curvature then vanishes, but the spatial Boyer–Lindquist coordinates are related to Cartesian coordinates by

$$\begin{aligned} x &= \sqrt{r^2 + a^2} \sin \theta \cos \phi, \\ y &= \sqrt{r^2 + a^2} \sin \theta \sin \phi, \\ z &= r \cos \theta. \end{aligned} \quad (6.18)$$

Eliminating θ gives

$$\frac{x^2 + y^2}{r^2 + a^2} + \frac{z^2}{r^2} = 1. \quad (6.19)$$

For fixed r , this is the equation of an oblate spheroid: the equatorial semi-axis is $\sqrt{r^2 + a^2}$ and the polar semi-axis is r . The surfaces $r = \text{constant}$ are therefore not spheres.

Computing $dx^2 + dy^2 + dz^2$, the mixed terms cancel and the flat metric becomes

$$ds^2 = -dt^2 + \frac{\Sigma}{r^2 + a^2} dr^2 + \Sigma d\theta^2 + (r^2 + a^2) \sin^2 \theta d\phi^2. \quad (6.20)$$

When $a \rightarrow 0$, Eq. (6.18) reduces to ordinary spherical coordinates and

$$\Sigma \rightarrow r^2. \quad (6.21)$$

The oblate geometry is therefore not an additional gravitational effect. It is already present in the coordinate system itself. When $M = Q = 0$, a survives only as a coordinate parameter and spacetime remains flat; when M or Q is non-zero and $J = Ma$, the same parameter also carries the physical information about rotation.

This distinction explains why r in the simulator is a coordinate radius, not an intrinsic Euclidean distance. Even in the flat limit, a circle drawn with numerical radius r does not automatically coincide with the equatorial section of the coordinate surface, whose Cartesian radius is $\sqrt{r^2 + a^2}$.

6.3 Why they are so useful

In the exterior region one may take

$$t \in \mathbb{R}, \quad r > r_+, \quad 0 \leq \theta \leq \pi, \quad \phi \sim \phi + 2\pi. \quad (6.22)$$

In this chart, the symmetry coordinates are directly t and ϕ . Boyer–Lindquist coordinates

- make the Killing fields ∂_t and ∂_ϕ manifest;
- isolate the genuine rotational mixing in the single t – ϕ term;
- separate the Hamilton–Jacobi equation for geodesics;
- make it easy to read off $E = -p_t$, $L_z = p_\phi$, the horizons $\Delta = 0$, and the static limit $g_{tt} = 0$;
- are asymptotically almost spherical and make M , Q , and J transparent.

They nevertheless have a decisive limitation: coefficients containing $1/\Delta$ diverge on the horizons. Crossing them requires ingoing/outgoing Kerr or Kerr–Schild coordinates. Boyer–Lindquist is therefore the natural chart for analysing the exterior and constants of motion, not for numerically following a trajectory through the horizon.

Chart	Advantage	Limitation
Boyer–Lindquist	Symmetries, constants of motion, separability	Singular at $\Delta = 0$
Ingoing Kerr	Regular on the future horizon	More off-diagonal terms
Kerr–Schild	Manifest flat background, horizon penetrating	Orbital geometry less immediate

The solver’s opening comment explicitly states: Boyer–Lindquist coordinates, signature $(-+++)$, $M = 1$. The check $r_0 > r_+ + 0.12M$ and stopping near r_+ are consequences of the chosen chart, not of the physical geodesic.

The Kerr–Newman metric and its limiting cases

The metric studied in this chapter is not introduced as a new ansatz. It is precisely the Kerr–Newman geometry already encountered in the preceding construction, now written in the Boyer–Lindquist coordinates obtained by applying Eq. (6.12) to the advanced metric (6.3).

The purpose of this chapter is therefore to analyse the resulting metric: first its compact algebraic form, then its individual components, inverse, electromagnetic field, and limiting cases.

7.1 What the solution must contain

Before writing the result explicitly, it is useful to recall the physical and geometrical requirements that identify the solution. Kerr–Newman must be stationary, axisymmetric, asymptotically flat, and satisfy the Einstein–Maxwell equations. It depends on the three global charges

$$M, \quad Q, \quad J = Ma, \quad (7.1)$$

and must reduce to the previously known solutions when one or more parameters vanish.

The following notation separates three geometrical roles:

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2, \quad \mathcal{A} = (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta. \quad (7.2)$$

These combinations are not additional assumptions. Σ and Δ have already appeared naturally in the rotating solution and in the Boyer–Lindquist transformation. The quantity $\mathcal{A}$ is introduced only as an abbreviation for the azimuthal coefficient.

As will become explicit below, Σ controls much of the angular dependence and its vanishing identifies the ring singularity, while Δ controls the radial structure and the locations of the horizons.

7.2 Compact form

After substituting the Boyer–Lindquist transformation (6.12) into the advanced metric (6.3), the radial cross terms cancel and the resulting metric can be rearranged algebraically into the compact form

$$\boxed{ds^2 = -\frac{\Delta}{\Sigma}(dt - a \sin^2 \theta d\phi)^2 + \frac{\sin^2 \theta}{\Sigma}[(r^2 + a^2)d\phi - a dt]^2 + \frac{\Sigma}{\Delta}dr^2 + \Sigma d\theta^2.} \quad (7.3)$$

Thus Eq. (7.3) is not guessed independently: it is a compact reorganization of the Boyer–Lindquist form of the already verified Kerr–Newman solution.

The squared form immediately shows that no terms

$$dt dr, \quad dr d\phi, \quad dt d\theta, \quad d\theta d\phi \quad (7.4)$$

occur. The only off-diagonal coupling is between the two Killing directions, t and ϕ . It also shows that the radial coefficient is Σ/Δ , anticipating the special role of the zeros of Δ .

7.3 Expanded form: the one implemented in the code

The simulator does not use the squares in Eq. (7.3) directly: it constructs the matrix $g_{\mu\nu}$. Expanding the two squares and collecting the coefficients of dt^2 , $dt d\phi$, and $d\phi^2$ gives

$$\begin{aligned} ds^2 = & - \left(1 - \frac{2Mr - Q^2}{\Sigma} \right) dt^2 - \frac{2a(2Mr - Q^2) \sin^2 \theta}{\Sigma} dt d\phi \\ & + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 \\ & + \left(r^2 + a^2 + \frac{a^2(2Mr - Q^2) \sin^2 \theta}{\Sigma} \right) \sin^2 \theta d\phi^2. \end{aligned} \quad (7.5)$$

Recall that in

$$g_{\mu\nu} dx^\mu dx^\nu, \quad (7.6)$$

the contributions $(\mu, \nu) = (t, \phi)$ and (ϕ, t) coincide. The coefficient written in front of $dt d\phi$ is therefore $2g_{t\phi}$, not $g_{t\phi}$.

The nonzero components are

$$\begin{aligned} g_{tt} &= - \left(1 - \frac{2Mr - Q^2}{\Sigma} \right), \\ g_{t\phi} &= - \frac{a(2Mr - Q^2) \sin^2 \theta}{\Sigma}, \\ g_{rr} &= \frac{\Sigma}{\Delta}, \quad g_{\theta\theta} = \Sigma, \\ g_{\phi\phi} &= \frac{\mathcal{A} \sin^2 \theta}{\Sigma}. \end{aligned} \quad (7.7)$$

The component $g_{t\phi}$ is precisely the surviving mixing between the stationary and axial Killing directions discussed in the previous chapter. Unlike the artificial cross term produced by a rotating coordinate chart on Schwarzschild, its asymptotic coefficient contains the invariant angular momentum $J = Ma$.

7.4 Inverse metric and determinant

The radial–polar block is already diagonal. For the t – ϕ block, use the inverse of the corresponding 2×2 matrix. The identity

$$g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -\Delta \sin^2 \theta \quad (7.8)$$

gives

$$\begin{aligned} g^{tt} &= - \frac{\mathcal{A}}{\Sigma \Delta}, \\ g^{t\phi} &= - \frac{a(2Mr - Q^2)}{\Sigma \Delta}, \\ g^{\phi\phi} &= \frac{\Delta - a^2 \sin^2 \theta}{\Sigma \Delta \sin^2 \theta}, \\ g^{rr} &= \frac{\Delta}{\Sigma}, \quad g^{\theta\theta} = \frac{1}{\Sigma}. \end{aligned} \quad (7.9)$$

The determinant takes the remarkably simple form

$$g = -\Sigma^2 \sin^2 \theta. \quad (7.10)$$

The absence of any factor $1/\Delta$ in the determinant is consistent with the fact that the divergence of Boyer–Lindquist components at $\Delta = 0$ is coordinate rather than intrinsic. Regular horizon-penetrating coordinates, or equivalently the behaviour of curvature invariants, are required for the fully coordinate-independent statement.

The vanishing of the determinant at $\sin \theta = 0$ instead has the same origin as the degeneration of ordinary polar coordinates on their axis and does not represent a curvature singularity.

7.5 The Kerr–Newman electromagnetic field

The Kerr–Newman solution is not specified by the metric alone. Since it is an electrovacuum solution of the Einstein–Maxwell system, the geometry must be accompanied by an electromagnetic field $F_{\mu\nu}$ satisfying

$$\nabla_\mu F^{\mu\nu} = 0, \quad (7.11)$$

while its electromagnetic stress–energy tensor acts as the source of the gravitational field through

$$G_{\mu\nu} = 8\pi T_{\mu\nu}^{\text{EM}}. \quad (7.12)$$

Stationarity and axisymmetry constrain the electromagnetic field in the same way that they constrain the geometry. In a gauge adapted to these symmetries, the potential may be chosen such that

$$\partial_t A_\mu = 0, \quad \partial_\phi A_\mu = 0, \quad (7.13)$$

so that its components depend only on r and θ .

These symmetry requirements do not determine the explicit functional form of A_μ . That form is part of the full Kerr–Newman solution of the coupled Einstein–Maxwell equations. In Boyer–Lindquist coordinates, a convenient gauge for this solution is

$$A_t = -\frac{Qr}{\Sigma}, \quad A_\phi = \frac{Qra \sin^2 \theta}{\Sigma}, \quad (7.14)$$

with

$$A_r = A_\theta = 0. \quad (7.15)$$

The choice of potential is not unique. Under the gauge transformation

$$A_\mu \longrightarrow A'_\mu = A_\mu + \partial_\mu \Lambda, \quad (7.16)$$

the electromagnetic tensor remains unchanged. The physically relevant field is therefore

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (7.17)$$

The form (7.14) has the correct non-rotating and asymptotic limits. When $a \rightarrow 0$, one has $\Sigma \rightarrow r^2$ and therefore

$$A_t \rightarrow -\frac{Q}{r}, \quad A_\phi \rightarrow 0, \quad (7.18)$$

recovering the Reissner–Nordström potential.

At large radius,

$$A_t = -\frac{Q}{r} + \mathcal{O}(r^{-3}), \quad A_\phi = \frac{Qa \sin^2 \theta}{r} + \mathcal{O}(r^{-3}). \quad (7.19)$$

The first asymptotic term identifies Q as the electric charge, whereas the azimuthal contribution is the leading term associated with the magnetic dipole generated by the rotating charged geometry. Its magnetic dipole moment is

$$\mu = Qa. \quad (7.20)$$

Using

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (7.21)$$

the independent nonzero components are

$$\begin{aligned} F_{tr} &= -\frac{Q(r^2 - a^2 \cos^2 \theta)}{\Sigma^2}, & F_{t\theta} &= \frac{2Qra^2 \sin \theta \cos \theta}{\Sigma^2}, \\ F_{r\phi} &= -a \sin^2 \theta \frac{Q(r^2 - a^2 \cos^2 \theta)}{\Sigma^2}, & F_{\theta\phi} &= \frac{2Qar(r^2 + a^2) \sin \theta \cos \theta}{\Sigma^2}. \end{aligned} \quad (7.22)$$

For $a = 0$, only

$$F_{tr} = -\frac{Q}{r^2} \quad (7.23)$$

survives, and the Reissner–Nordström electromagnetic field is recovered.

The potential (7.14) is therefore not determined by symmetry alone. Rather, it is a convenient gauge representation of the electromagnetic field belonging to the Kerr–Newman solution. Together, the metric (7.5) and the field $F_{\mu\nu}$ above satisfy the coupled Einstein–Maxwell equations.

The current simulator uses Q to modify the geometry, but integrates geodesics of neutral particles and therefore does not add a Lorentz-force term constructed from $F_{\mu\nu}$.

7.6 Checks in the four limits

The limiting cases provide important consistency checks of the Kerr–Newman solution. Recall that, in Boyer–Lindquist coordinates, the metric can be written as

$$\boxed{ds^2 = -\frac{\Delta}{\Sigma} (dt - a \sin^2 \theta d\phi)^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \frac{\sin^2 \theta}{\Sigma} [(r^2 + a^2)d\phi - a dt]^2}, \quad (7.24)$$

where

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2. \quad (7.25)$$

The parameter a is the specific angular momentum of the black hole. In the geometrized units used here,

$$\boxed{a = \frac{J}{M}, \quad J = Ma,} \quad (7.26)$$

where J is the total angular momentum measured at infinity. Therefore $a \neq 0$ represents physical rotation only when it is associated with a nonzero angular momentum J of the spacetime.

The four principal limits are

Choice	a	Q	Solution
no rotation, no charge	0	0	Schwarzschild
charge without rotation	0	$\neq 0$	Reissner–Nordström
rotation without charge	$\neq 0$	0	Kerr
rotation and charge	$\neq 0$	$\neq 0$	Kerr–Newman

For $a = 0$, one has

$$\Sigma = r^2, \quad \Delta = r^2 - 2Mr + Q^2, \quad (7.27)$$

and the metric reduces to

$$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\Omega^2, \quad (7.28)$$

which is the Reissner–Nordström metric. Setting also $Q = 0$ gives Schwarzschild.

Conversely, setting $Q = 0$ while retaining $a \neq 0$ gives

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2, \quad (7.29)$$

and the Kerr metric is recovered. Thus Kerr–Newman continuously connects the static charged solution to the rotating uncharged solution.

There is also a purely coordinate check. Setting

$$M = Q = 0 \quad (7.30)$$

while formally retaining $a \neq 0$, the curvature vanishes and the metric reduces to Minkowski spacetime written in the oblate spheroidal coordinates introduced in the preceding chapter.

This limit is particularly useful for distinguishing the coordinate parameter a from physical angular momentum. Since

$$J = Ma, \quad (7.31)$$

when $M = 0$ the physical angular momentum is

$$J = 0, \quad (7.32)$$

even if the coordinate parameter a is retained. In this limit, therefore, a describes only the oblate spheroidal coordinate system and not a rotating gravitational source.

The physical relation between a and the angular momentum becomes especially transparent in the asymptotic form of the metric. For $r \rightarrow \infty$,

$$\begin{aligned} g_{tt} &= -1 + \frac{2M}{r} - \frac{Q^2}{r^2} + \mathcal{O}(r^{-3}), \\ g_{t\phi} &= -\frac{2Ma \sin^2 \theta}{r} + \mathcal{O}(r^{-2}) \\ &= -\frac{2J \sin^2 \theta}{r} + \mathcal{O}(r^{-2}), \\ A_t &= -\frac{Q}{r} + \mathcal{O}(r^{-3}). \end{aligned} \quad (7.33)$$

The coefficient of the leading $1/r$ term in g_{tt} identifies the mass M , while the Coulomb term in A_t identifies the electric charge Q . The leading rotational contribution to $g_{t\phi}$ is proportional to the total angular momentum:

$$\boxed{g_{t\phi} \sim -\frac{2J \sin^2 \theta}{r}}. \quad (7.34)$$

Consequently, comparison with the Kerr–Newman component

$$g_{t\phi} = -\frac{a(2Mr - Q^2) \sin^2 \theta}{\Sigma} \quad (7.35)$$

gives, at leading order,

$$J = Ma. \quad (7.36)$$

Thus the three parameters appearing in the Kerr–Newman solution have a direct asymptotic interpretation as the global mass, electric charge, and angular momentum:

$$\boxed{M, \quad Q, \quad J = Ma.} \quad (7.37)$$

7.7 SI units in the function Δ

If the coordinates (t, r, a) have the dimension of length,

$$\Delta = r^2 - 2\frac{GM_{\text{SI}}}{c^2}r + \left(\frac{J_{\text{SI}}}{M_{\text{SI}}c}\right)^2 + \frac{GQ_{\text{SI}}^2}{4\pi\epsilon_0 c^4}. \quad (7.38)$$

Writing Q^2 and a^2 alongside r^2 is legitimate only after these conversions. The simulator performs them before normalizing $M = 1$.

Horizons, ergoregion, and the rotating singularity

8.1 The horizons: $\Delta = 0$

As in the spherical case, consider the hypersurface

$$r = \text{constant}. \quad (8.1)$$

It can be described as a level surface of the scalar function

$$\Phi(x) = r - r_0, \quad (8.2)$$

whose normal is

$$n_\mu = \partial_\mu \Phi = \partial_\mu r. \quad (8.3)$$

Since $\partial_\mu r$ has only a radial component, its squared norm is

$$n_\mu n^\mu = g^{\mu\nu} n_\mu n_\nu = g^{rr} = \frac{\Delta}{\Sigma}. \quad (8.4)$$

Away from the curvature singularity, where $\Sigma > 0$, the hypersurface $r = \text{constant}$ is therefore null if and only if

$$\Delta = 0. \quad (8.5)$$

Since

$$\Delta = r^2 - 2Mr + a^2 + Q^2, \quad (8.6)$$

the two roots are

$$r_\pm = M \pm \sqrt{M^2 - a^2 - Q^2}. \quad (8.7)$$

The calculation above establishes locally that the hypersurfaces $r = r_\pm$ are null. The identification of the outer root r_+ with the event horizon is instead a global property of the Kerr–Newman spacetime.

There are three regimes:

$$\begin{aligned} M^2 > a^2 + Q^2 : & \quad r_- < r_+ \quad \text{subextremal,} \\ M^2 = a^2 + Q^2 : & \quad r_- = r_+ = M \quad \text{extremal,} \\ M^2 < a^2 + Q^2 : & \quad \text{no real horizon.} \end{aligned} \quad (8.8)$$

In the subextremal case, the outer root r_+ is the event horizon associated with the asymptotically flat exterior region. The inner root r_- is a Cauchy horizon and inherits the strong blueshift instability already encountered in Reissner–Nordström. For $Q = 0$, the Kerr horizons are recovered, while for $a = 0$ the Reissner–Nordström horizons are recovered.

The rotating character of the horizon requires one additional step. The Kerr–Newman geometry is stationary and axisymmetric and therefore possesses the two Killing fields

$$\xi = \partial_t, \quad \eta = \partial_\phi. \quad (8.9)$$

The first generates time translations, while the second generates rotations about the symmetry axis.

For a non-rotating black hole, the horizon generator can coincide with the stationary Killing field ∂_t . In Kerr–Newman this is no longer true: the event horizon itself rotates. Its null generator is instead a linear combination of the stationary and axial Killing fields,

$$\boxed{\chi = \xi + \Omega_H \eta = \partial_t + \Omega_H \partial_\phi}, \quad (8.10)$$

where Ω_H is the angular velocity of the horizon.

To determine Ω_H , recall that the determinant of the t – ϕ block of the metric satisfies

$$g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -\Delta \sin^2 \theta. \quad (8.11)$$

At the outer horizon,

$$\Delta(r_+) = 0, \quad (8.12)$$

and therefore

$$g_{tt}g_{\phi\phi} - g_{t\phi}^2 = 0. \quad (8.13)$$

The t – ϕ block consequently becomes degenerate and possesses a non-trivial null direction.

Since χ contains only the t and ϕ components, its components can be written as

$$\chi^\mu = (1, 0, 0, \Omega_H). \quad (8.14)$$

The degenerate direction of the t – ϕ block satisfies

$$\begin{pmatrix} g_{tt} & g_{t\phi} \\ g_{t\phi} & g_{\phi\phi} \end{pmatrix} \begin{pmatrix} 1 \\ \Omega_H \end{pmatrix} = 0. \quad (8.15)$$

In particular,

$$g_{t\phi} + \Omega_H g_{\phi\phi} = 0, \quad (8.16)$$

so that

$$\Omega_H = - \left. \frac{g_{t\phi}}{g_{\phi\phi}} \right|_{r=r_+}. \quad (8.17)$$

For Kerr–Newman,

$$g_{t\phi} = - \frac{a(2Mr - Q^2) \sin^2 \theta}{\Sigma}, \quad (8.18)$$

while

$$g_{\phi\phi} = \frac{\mathcal{A} \sin^2 \theta}{\Sigma}, \quad (8.19)$$

with

$$\mathcal{A} = (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta. \quad (8.20)$$

At $r = r_+$, the horizon equation

$$\Delta(r_+) = r_+^2 - 2Mr_+ + a^2 + Q^2 = 0 \quad (8.21)$$

implies

$$2Mr_+ - Q^2 = r_+^2 + a^2. \quad (8.22)$$

Moreover,

$$\mathcal{A}(r_+) = (r_+^2 + a^2)^2. \quad (8.23)$$

Therefore

$$\begin{aligned}
 \Omega_H &= - \left. \frac{g_{t\phi}}{g_{\phi\phi}} \right|_{r=r_+} \\
 &= \frac{a(r_+^2 + a^2) \sin^2 \theta / \Sigma_+}{(r_+^2 + a^2)^2 \sin^2 \theta / \Sigma_+} \\
 &= \boxed{\frac{a}{r_+^2 + a^2}}.
 \end{aligned} \tag{8.24}$$

The dependence on θ cancels completely. Hence the entire horizon is generated with the same angular velocity Ω_H .

Finally, we can explicitly verify that χ is null on the horizon. Its norm is

$$\chi^\mu \chi_\mu = g_{tt} + 2\Omega_H g_{t\phi} + \Omega_H^2 g_{\phi\phi}. \tag{8.25}$$

Using

$$\Omega_H = - \frac{g_{t\phi}}{g_{\phi\phi}}, \tag{8.26}$$

one obtains

$$\begin{aligned}
 \chi^\mu \chi_\mu &= g_{tt} - \frac{g_{t\phi}^2}{g_{\phi\phi}} \\
 &= \frac{g_{tt}g_{\phi\phi} - g_{t\phi}^2}{g_{\phi\phi}} \\
 &= - \frac{\Delta \sin^2 \theta}{g_{\phi\phi}}.
 \end{aligned} \tag{8.27}$$

Consequently,

$$\boxed{\chi^\mu \chi_\mu = 0 \quad (r = r_+).} \tag{8.28}$$

The event horizon is therefore generated not by a pure time translation, but by a time translation accompanied by an azimuthal rotation. Along an integral curve of the horizon generator,

$$\frac{dt}{d\lambda} = 1, \quad \frac{d\phi}{d\lambda} = \Omega_H, \tag{8.29}$$

and hence

$$\boxed{\frac{d\phi}{dt} = \Omega_H.} \tag{8.30}$$

Thus Ω_H has the direct physical interpretation of the angular velocity of the Kerr–Newman event horizon.

8.2 Static limit: $g_{tt} = 0$

A worldline with constant spatial Boyer–Lindquist coordinates (r, θ, ϕ) has tangent vector proportional to the stationary Killing field

$$\xi = \partial_t. \tag{8.31}$$

Such a worldline represents an observer remaining at rest with respect to the asymptotically non-rotating Boyer–Lindquist frame. With the signature $(-+++)$, this is physically possible only where ∂_t is timelike, that is,

$$g_{tt} < 0. \tag{8.32}$$

For Kerr–Newman,

$$g_{tt} = - \left(1 - \frac{2Mr - Q^2}{\Sigma} \right), \quad \Sigma = r^2 + a^2 \cos^2 \theta. \quad (8.33)$$

The boundary beyond which a static worldline ceases to be timelike is therefore determined by

$$g_{tt} = 0. \quad (8.34)$$

Explicitly,

$$\begin{aligned} 1 - \frac{2Mr - Q^2}{\Sigma} &= 0, \\ \Sigma - 2Mr + Q^2 &= 0, \\ r^2 - 2Mr + a^2 \cos^2 \theta + Q^2 &= 0. \end{aligned} \quad (8.35)$$

The two roots are

$$\boxed{r_s^\pm(\theta) = M \pm \sqrt{M^2 - Q^2 - a^2 \cos^2 \theta}.} \quad (8.36)$$

The outer surface

$$r = r_s^+(\theta) \quad (8.37)$$

is called the *outer stationary-limit surface*, or outer *ergosurface*. It is important to distinguish the surface from the region that it bounds: the outer stationary-limit surface is the outer boundary of the *ergoregion* (or *ergosphere*), whose inner boundary is the outer event horizon $r = r_+$. The ergoregion itself will be analysed in the following section.

Unlike the event horizon, whose Boyer–Lindquist radius

$$r_+ = M + \sqrt{M^2 - a^2 - Q^2} \quad (8.38)$$

is independent of θ , the stationary-limit surface depends on latitude. It is therefore not a surface of constant Boyer–Lindquist radius.

Since

$$\cos^2 \theta \leq 1, \quad (8.39)$$

one has

$$M^2 - Q^2 - a^2 \cos^2 \theta \geq M^2 - Q^2 - a^2, \quad (8.40)$$

and consequently

$$\boxed{r_s^+(\theta) \geq r_+}. \quad (8.41)$$

At the rotation axis, $\theta = 0, \pi$, one has $\cos^2 \theta = 1$, and therefore

$$r_s^+(0) = r_s^+(\pi) = r_+. \quad (8.42)$$

Thus the outer ergosurface touches the event horizon at the two poles.

At the equatorial plane,

$$\theta = \frac{\pi}{2}, \quad \cos \theta = 0, \quad (8.43)$$

so the outer stationary limit becomes

$$\boxed{r_s^+ \left(\frac{\pi}{2} \right) = M + \sqrt{M^2 - Q^2}.} \quad (8.44)$$

This is the *equatorial static limit*. Notice that this particular radius contains no explicit dependence on a . This does not mean that rotation is irrelevant to the equatorial ergoregion: the outer horizon

$$r_+ = M + \sqrt{M^2 - a^2 - Q^2} \quad (8.45)$$

does depend on a . Therefore the equatorial width of the ergoregion,

$$r_s^+ \left(\frac{\pi}{2} \right) - r_+, \quad (8.46)$$

increases as the magnitude of the spin increases, for fixed M and Q .

This is precisely the quantity displayed by the current simulator as the *equatorial static limit*. The Web Demo evolves trajectories with equatorial initial conditions and visualizes an equatorial two-dimensional section. Consequently, it uses the single radius $r_s^+(\pi/2)$ rather than the complete θ -dependent stationary-limit surface.

For intuition

Imagine an observer approaching the rotating black hole from far away. At large distance, the observer may in principle use acceleration to remain at fixed Boyer–Lindquist coordinates and therefore remain static with respect to infinity.

As the black hole is approached, frame dragging becomes increasingly strong: the local inertial frames are progressively dragged in the azimuthal direction. The outer stationary-limit surface marks the point at which remaining exactly static with respect to infinity ceases to be physically possible.

Nothing material is located on this surface. A freely falling observer does not encounter a wall, a membrane, or a curvature singularity when crossing it, and therefore does not “see” an ergosurface as a physical object. Rather, the surface marks a change in the possible causal motions of observers.

Crossing the outer stationary-limit surface means entering the ergoregion. The precise causal restriction on the observer’s angular motion, and its relation to frame dragging, are derived in the following section.

8.3 Ergoregion and inertial-frame dragging

As anticipated above, the *ergoregion*, also commonly called the *ergosphere*, is the region lying between the outer event horizon and the outer stationary-limit surface:

$$r_+ < r < r_s^+(\theta). \quad (8.47)$$

Thus the outer stationary-limit surface $r = r_s^+(\theta)$ is the outer boundary of the ergoregion, while $r = r_+$ is its inner boundary. At the rotation axis the two surfaces meet, so the ergoregion shrinks to zero width at the poles.

Inside the ergoregion one has

$$g_{tt} > 0. \quad (8.48)$$

Since

$$g(\partial_t, \partial_t) = g_{tt}, \quad (8.49)$$

the stationary Killing vector ∂_t is spacelike there. Consequently, a worldline with

$$r, \theta, \phi = \text{constant} \quad (8.50)$$

would be spacelike and therefore cannot describe a physical observer.

This statement should not be confused with the causal behaviour inside the event horizon. An observer inside the ergoregion but outside r_+ is not forced to fall into the black hole: radial outward motion is still possible. What is forbidden is remaining at rest with respect to infinity. Every timelike or null observer must acquire an azimuthal motion associated with the rotation of the spacetime.

To make this statement quantitative, consider at a given event a causal trajectory with fixed instantaneous r and θ and define its angular velocity by

$$\Omega = \frac{d\phi}{dt}. \quad (8.51)$$

Its tangent direction is proportional to

$$\partial_t + \Omega \partial_\phi. \quad (8.52)$$

Along such a direction,

$$\frac{ds^2}{dt^2} = g_{tt} + 2g_{t\phi}\Omega + g_{\phi\phi}\Omega^2. \quad (8.53)$$

Causality requires

$$g_{tt} + 2g_{t\phi}\Omega + g_{\phi\phi}\Omega^2 \leq 0. \quad (8.54)$$

The two null limiting angular velocities are therefore the roots of

$$g_{\phi\phi}\Omega^2 + 2g_{t\phi}\Omega + g_{tt} = 0, \quad (8.55)$$

namely

$$\Omega_{\pm} = \frac{-g_{t\phi} \pm \sqrt{g_{t\phi}^2 - g_{tt}g_{\phi\phi}}}{g_{\phi\phi}}. \quad (8.56)$$

A timelike observer must have an angular velocity between these two null values.

Using the identity

$$g_{t\phi}^2 - g_{tt}g_{\phi\phi} = \Delta \sin^2 \theta, \quad (8.57)$$

one obtains

$$\Omega_{\pm} = \frac{-g_{t\phi} \pm \sqrt{\Delta} \sin \theta}{g_{\phi\phi}} \quad (8.58)$$

in the exterior region.

At the outer stationary-limit surface $g_{tt} = 0$, one of the two limiting angular velocities becomes zero. After entering the ergoregion, $g_{tt} > 0$, and $\Omega = 0$ no longer satisfies the causal inequality. Therefore remaining static is impossible.

For $a > 0$, both allowed limiting angular velocities become positive inside the ergoregion, so every future-directed causal observer must rotate in the same azimuthal sense as the black hole. For $a < 0$, the corresponding signs are reversed. This compulsory azimuthal motion is the strong-field manifestation of inertial-frame dragging.

A particularly useful family of observers is given by the *zero-angular-momentum observers* (ZAMOs). For an observer with angular velocity

$$\Omega = \frac{u^\phi}{u^t}, \quad (8.59)$$

the conserved axial angular momentum per unit mass is

$$L_z = u_\phi = g_{t\phi}u^t + g_{\phi\phi}u^\phi. \quad (8.60)$$

Imposing

$$L_z = 0 \quad (8.61)$$

gives

$$g_{t\phi} + g_{\phi\phi}\Omega = 0. \quad (8.62)$$

Hence the ZAMO angular velocity is

$$\omega = -\frac{g_{t\phi}}{g_{\phi\phi}} = \frac{a(2Mr - Q^2)}{\mathcal{A}}. \quad (8.63)$$

Thus even an observer carrying zero conserved angular momentum relative to infinity generally has

$$\frac{d\phi}{dt} = \omega \neq 0. \quad (8.64)$$

This is a direct manifestation of frame dragging: the local inertial frames themselves are dragged by the rotation of the black hole.

Finally, approaching the outer horizon from the exterior,

$$\Delta \longrightarrow 0, \quad (8.65)$$

so the two limiting angular velocities merge:

$$\Omega_+ \longrightarrow \Omega_H, \quad \Omega_- \longrightarrow \Omega_H. \quad (8.66)$$

At the same time,

$$\omega \longrightarrow \Omega_H = \frac{a}{r_+^2 + a^2}. \quad (8.67)$$

Therefore every causal direction tangent to constant r and θ is forced towards the angular velocity of the horizon itself.

For intuition

The outer stationary-limit surface and the event horizon have fundamentally different meanings. The former is the outer boundary of the ergoregion and marks where static observers cease to exist; it is not itself a causal barrier and can be crossed in both directions. The event horizon is instead a causal boundary: once it is crossed towards the future, no future-directed causal trajectory can return to the original asymptotically flat exterior region.

8.4 Ring singularity

The true curvature singularity lies where

$$\Sigma = r^2 + a^2 \cos^2 \theta = 0. \quad (8.68)$$

Since both terms are non-negative, for $a \neq 0$ this can occur only if

$$r = 0, \quad \theta = \frac{\pi}{2}. \quad (8.69)$$

This condition is fundamentally different from $\Delta = 0$. The surfaces $\Delta = 0$ correspond to horizons, where some Boyer–Lindquist metric components become singular because the coordinate chart degenerates, while curvature invariants remain finite. By contrast, when $\Sigma \rightarrow 0$ curvature invariants

constructed from the Riemann tensor diverge. Therefore $\Sigma = 0$ identifies an intrinsic singularity of the spacetime geometry, which cannot be removed by a change of coordinates.

In the oblate spheroidal coordinates (6.18), setting $r = 0$ gives

$$z = 0, \quad x^2 + y^2 = a^2 \sin^2 \theta. \quad (8.70)$$

As θ varies, this covers a disk of radius $|a|$ in the equatorial plane. However, the singularity occurs only when $\theta = \pi/2$, for which

$$x^2 + y^2 = a^2, \quad z = 0. \quad (8.71)$$

Thus not all of $r = 0$ is singular: the singular set is only the rim of this disk, namely the ring

$$\boxed{x^2 + y^2 = a^2, \quad z = 0.} \quad (8.72)$$

The ideal analytic extension permits regions with $r < 0$ and contains closed timelike curves near the ring. As with the chain of Reissner–Nordström regions, these global properties of the eternal exact solution must not be identified without caution with the interior produced by realistic gravitational collapse [1, 9].

8.5 Area, surface gravity, and horizon potential

To derive the area, consider a spatial section of the outer horizon $r = r_+$. Since $\Delta(r_+) = 0$,

$$g_{\theta\theta}|_{r_+} = \Sigma_+, \quad g_{\phi\phi}|_{r_+} = \frac{(r_+^2 + a^2)^2 \sin^2 \theta}{\Sigma_+}, \quad (8.73)$$

where

$$\Sigma_+ = r_+^2 + a^2 \cos^2 \theta. \quad (8.74)$$

The induced area element on a constant-time spatial section of the horizon is therefore

$$\begin{aligned} dA &= \sqrt{g_{\theta\theta}g_{\phi\phi}} d\theta d\phi \\ &= (r_+^2 + a^2) \sin \theta d\theta d\phi. \end{aligned} \quad (8.75)$$

Integrating over $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$ gives

$$\boxed{A_H = 4\pi(r_+^2 + a^2).} \quad (8.76)$$

The other principal quantities associated with the outer horizon are

$$\kappa = \frac{r_+ - r_-}{2(r_+^2 + a^2)}, \quad (8.77)$$

$$\begin{aligned} \Omega_H &= \frac{a}{r_+^2 + a^2}, \\ \Phi_H &= \frac{Qr_+}{r_+^2 + a^2}. \end{aligned} \quad (8.78)$$

The surface gravity κ characterizes the non-affine parametrization of the null generators of the Killing horizon. If

$$\chi = \partial_t + \Omega_H \partial_\phi \quad (8.79)$$

is the horizon-generating Killing field, then on the horizon

$$\chi^\nu \nabla_\nu \chi^\mu = \kappa \chi^\mu. \quad (8.80)$$

Thus κ measures how the natural Killing parameter along the horizon differs from an affine parameter along its null generators.

The horizon electrostatic potential is defined by

$$\Phi_H = -\chi^\mu A_\mu|_{r_+}. \quad (8.81)$$

Using

$$\chi = \partial_t + \Omega_H \partial_\phi, \quad (8.82)$$

one has

$$\Phi_H = -(A_t + \Omega_H A_\phi)|_{r_+}. \quad (8.83)$$

Substituting the Kerr–Newman potential and

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad (8.84)$$

the dependence on θ cancels and one obtains

$$\boxed{\Phi_H = \frac{Qr_+}{r_+^2 + a^2}.} \quad (8.85)$$

Both Φ_H and κ are constant over the horizon, an essential property in black-hole mechanics.

The horizon quantities satisfy the classical first law

$$\boxed{dM = \frac{\kappa}{8\pi} dA_H + \Omega_H dJ + \Phi_H dQ.} \quad (8.86)$$

Here the differentials do not represent derivatives along the trajectory of a particle or the time evolution of a single stationary black hole. Instead, they describe infinitesimal variations between nearby stationary Kerr–Newman solutions. In other words, one compares a black hole with parameters (M, J, Q) to a nearby one with

$$(M + dM, J + dJ, Q + dQ). \quad (8.87)$$

The first law states that the corresponding change in the total mass-energy can be decomposed into three contributions:

$$\frac{\kappa}{8\pi} dA_H, \quad \Omega_H dJ, \quad \Phi_H dQ. \quad (8.88)$$

They are respectively associated with a change in horizon area, angular momentum, and electric charge.

In the extremal limit,

$$r_+ = r_-, \quad (8.89)$$

one has

$$\kappa = 0. \quad (8.90)$$

Black-hole thermodynamics is not required by the solver, but these relations show that the same parameters M , J , and Q used by the simulator also control the mechanics of the Kerr–Newman horizon.

8.6 Negative energy and the Penrose process

Because the Kerr–Newman geometry is stationary, the vector field

$$\xi = \partial_t \quad (8.91)$$

is a Killing vector. A Killing symmetry generates a conserved quantity along geodesic motion. For a particle with four-momentum p^μ , the conserved quantity associated with ξ is

$$E = -p_\mu \xi^\mu. \quad (8.92)$$

In Boyer–Lindquist coordinates, since

$$\xi^\mu = (1, 0, 0, 0), \quad (8.93)$$

one immediately obtains

$$E = -p_t. \quad (8.94)$$

For intuition

The conservation of E follows directly from the Killing equation. If ξ^μ is a Killing vector,

$$\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0. \quad (8.95)$$

Along an affinely parametrized geodesic, the four-momentum is parallel transported,

$$p^\mu \nabla_\mu p_\nu = 0. \quad (8.96)$$

Therefore

$$\begin{aligned} p^\mu \nabla_\mu (p_\nu \xi^\nu) &= (p^\mu \nabla_\mu p_\nu) \xi^\nu + p^\mu p^\nu \nabla_\mu \xi_\nu \\ &= p^\mu p^\nu \nabla_\mu \xi_\nu. \end{aligned} \quad (8.97)$$

Since $p^\mu p^\nu$ is symmetric in μ, ν , only the symmetric part of $\nabla_\mu \xi_\nu$ contributes:

$$\begin{aligned} p^\mu p^\nu \nabla_\mu \xi_\nu &= \frac{1}{2} p^\mu p^\nu (\nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu) \\ &= 0. \end{aligned} \quad (8.98)$$

Hence

$$p_\mu \xi^\mu = \text{constant} \quad (8.99)$$

along the geodesic. With signature $(-+++)$, the conventional energy associated with the stationary Killing field is defined with the opposite sign,

$$E = -p_\mu \xi^\mu. \quad (8.100)$$

Far from the black hole, $\xi = \partial_t$ approaches the ordinary future-directed time-translation vector of Minkowski spacetime. The conserved quantity E can therefore be interpreted as the energy of the particle measured relative to infinity.

Outside the ergoregion, ξ is timelike and future-directed. For any future-directed causal four-momentum p^μ , one then has

$$p_\mu \xi^\mu < 0, \quad (8.101)$$

and consequently

$$E = -p_\mu \xi^\mu > 0. \quad (8.102)$$

Inside the ergoregion, however,

$$g(\xi, \xi) = g_{tt} > 0, \quad (8.103)$$

so the stationary Killing vector ξ becomes spacelike. The scalar product $p_\mu \xi^\mu$ is then no longer forced to be negative, and causal states with

$$\boxed{E < 0} \quad (8.104)$$

become possible.

This possibility is the essential ingredient of the Penrose process. Suppose that a body enters the ergoregion with conserved energy E_0 and splits into two fragments with energies E_1 and E_2 . Conservation of the Killing energy gives

$$E_0 = E_1 + E_2. \quad (8.105)$$

If one fragment enters the black hole on an allowed negative-energy orbit,

$$E_1 < 0, \quad (8.106)$$

then the other fragment can escape to infinity with

$$E_2 = E_0 - E_1 > E_0. \quad (8.107)$$

The escaping fragment therefore carries more energy than the original body.

The additional energy is not created: it is extracted from the rotational energy of the black hole. The fragment crossing the horizon with negative Killing energy reduces the black hole's mass and angular momentum, and the process therefore decreases its rotational energy and J .

On the equatorial plane, the canvas represents the outer horizon, the dashed inner horizon, and the equatorial static limit as circles with coordinate radii r_+ , r_- , and $r_s^+(\pi/2)$. For $a \neq 0$, this is a projection $x = r \cos \phi$, $y = r \sin \phi$, not a Euclidean embedding of the spatial geometry or a photograph of the black hole.

Kerr–Newman geodesics

9.1 Lagrangian and normalization

A test particle is sufficiently light not to modify M , Q , or J . If it is also neutral, its motion depends only on the metric and can be obtained from the Lagrangian

$$\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu, \quad (9.1)$$

where the dot denotes differentiation with respect to proper time for a massive particle or an affine parameter for a photon.

For intuition

Why this Lagrangian?

For a freely falling massive particle, the fundamental action is proportional to the proper time measured along its worldline:

$$S = -m \int d\tau. \quad (9.2)$$

^a Using the spacetime metric with signature $(-+++)$,

$$d\tau^2 = -g_{\mu\nu}dx^\mu dx^\nu, \quad (9.3)$$

so, for an arbitrary parameter λ along the worldline,

$$S = -m \int \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda. \quad (9.4)$$

The corresponding Lagrangian is therefore

$$\mathcal{L}_{\text{sqr}} = -m \sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}. \quad (9.5)$$

This action has a direct geometrical meaning: a freely falling massive particle extremizes the proper time between nearby events. However, after choosing an affine parameter along the geodesic, one may use the dynamically equivalent quadratic Lagrangian

$$\boxed{\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu.} \quad (9.6)$$

The overall factor $1/2$ has no physical significance; it is chosen because it simplifies the Euler–Lagrange equations.

To see the equivalence more explicitly, define

$$X = -g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu. \quad (9.7)$$

Then

$$\mathcal{L}_{\text{sqr}} = -m\sqrt{X}. \quad (9.8)$$

Its derivative with respect to $\dot{x}^\mu$ is

$$\frac{\partial \mathcal{L}_{\text{sqr}}}{\partial \dot{x}^\mu} = m \frac{g_{\mu\nu}\dot{x}^\nu}{\sqrt{X}}. \quad (9.9)$$

If λ is affine, and in particular if $\lambda = \tau$ for a massive particle, X is constant along the geodesic. The factor $m/\sqrt{X}$ is then constant and drops out of the Euler–Lagrange equations, leaving exactly the same trajectory obtained from

$$\mathcal{L} = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu. \quad (9.10)$$

For a massive particle parametrized by proper time,

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -1, \quad (9.11)$$

whereas for a photon the tangent vector is null,

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0. \quad (9.12)$$

The quadratic Lagrangian is therefore convenient because the same formalism describes both timelike and null geodesics.

^aThe form of the free-particle action is motivated by requiring locality, general covariance, the absence of external fields, and invariance under reparametrizations of the worldline parameter.

Since the norm of the tangent is conserved along an affinely parametrized geodesic, choose

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -\varepsilon, \quad \varepsilon = \begin{cases} 1, & \text{massive particle,} \\ 0, & \text{photon.} \end{cases} \quad (9.13)$$

The Euler–Lagrange equations are

$$\frac{d}{d\lambda}(g_{\mu\nu}\dot{x}^\nu) - \frac{1}{2}\partial_\mu g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0. \quad (9.14)$$

Raising the index with $g^{\mu\nu}$ and recognizing the Levi–Civita connection, they are equivalent to the geodesic equation

$$\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0. \quad (9.15)$$

9.2 Energy and axial angular momentum

The canonical momentum associated with the Lagrangian is

$$p_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = g_{\mu\nu}\dot{x}^\nu. \quad (9.16)$$

Since the metric depends on neither t nor ϕ , the corresponding Euler–Lagrange equations give $\dot{p}_t = 0$ and $\dot{p}_\phi = 0$. Therefore define

$$\boxed{E = -p_t, \quad L_z = p_\phi.} \quad (9.17)$$

Explicitly,

$$\begin{aligned} -E &= g_{tt}\dot{t} + g_{t\phi}\dot{\phi}, \\ L_z &= g_{t\phi}\dot{t} + g_{\phi\phi}\dot{\phi}. \end{aligned} \quad (9.18)$$

These two equations form a linear system for $\dot{t}$ and $\dot{\phi}$. Multiplying by the inverse of the t – ϕ block gives

$$\dot{t} = -g^{tt}E + g^{t\phi}L_z, \quad (9.19)$$

$$\dot{\phi} = -g^{t\phi}E + g^{\phi\phi}L_z. \quad (9.20)$$

These are exactly the formulas used by the simulator to construct the initial time and azimuthal components.

For a massive particle, E and L_z are specific quantities, namely divided by the rest mass. For a photon they depend on the scale of the affine parameter; the ratio

$$b = \frac{L_z}{E} \quad (9.21)$$

is the relevant impact parameter. With $M = 1$, the displayed value is b/M .

9.3 Carter constant and separability

The two manifest symmetries provide E and L_z , but are not by themselves enough to make generic motion completely integrable. Kerr–Newman also possesses a hidden symmetry described by a rank-two Killing tensor. It emerges upon separating the Hamilton–Jacobi equation

$$\frac{\partial S}{\partial \lambda} + \frac{1}{2} g^{\mu\nu} \partial_\mu S \partial_\nu S = 0 \quad (9.22)$$

with the ansatz

$$S = -Et + L_z \phi + S_r(r) + S_\theta(\theta) + \frac{1}{2} \varepsilon \lambda. \quad (9.23)$$

The r -dependent and θ -dependent terms must equal the same separation constant, the Carter constant $\mathcal{Q} = \Sigma^2 \dot{\theta}^2$. Define

$$P(r) = E(r^2 + a^2) - aL_z. \quad (9.24)$$

For a neutral particle, the first-order equations are

$$\Sigma^2 \dot{r}^2 = R(r) = P(r)^2 - \Delta \left[\varepsilon r^2 + (L_z - aE)^2 + \mathcal{Q} \right], \quad (9.25)$$

$$\Sigma^2 \dot{\theta}^2 = \Theta(\theta) = \mathcal{Q} - \cos^2 \theta \left[a^2 (\varepsilon - E^2) + \frac{L_z^2}{\sin^2 \theta} \right], \quad (9.26)$$

$$\Sigma \dot{t} = \frac{(r^2 + a^2)P}{\Delta} - a(aE \sin^2 \theta - L_z), \quad (9.27)$$

$$\Sigma \dot{\phi} = \frac{aP}{\Delta} - \left(aE - \frac{L_z}{\sin^2 \theta} \right). \quad (9.28)$$

For a charged particle with specific charge $e = q/m$, the action also contains $eA_\mu \dot{x}^\mu$ and the equation of motion becomes

$$u^\nu \nabla_\nu u^\mu = e F^\mu{}_\nu u^\nu. \quad (9.29)$$

The conserved quantities then involve the canonical momentum $u_\mu + eA_\mu$, and $P(r)$ contains the additional term $-eQr$. The motion is not a geodesic of the metric alone; for this reason, the equations above and the simulator explicitly refer to neutral particles.

9.4 Why the equatorial plane remains equatorial

The current Web Demo restricts the initial conditions to the equatorial plane,

$$\theta_0 = \frac{\pi}{2}, \quad \dot{\theta}_0 = 0. \quad (9.30)$$

This restriction is not an artificial constraint imposed during the subsequent evolution. It is consistent with an exact symmetry of the Kerr–Newman geometry.

Indeed, the metric is invariant under reflection across the equatorial plane,

$$\theta \mapsto \pi - \theta. \quad (9.31)$$

The surface

$$\theta = \frac{\pi}{2} \quad (9.32)$$

is the fixed-point set of this reflection. Consider a geodesic whose initial point lies on this surface and whose initial tangent is tangent to it, so that

$$\theta_0 = \frac{\pi}{2}, \quad \dot{\theta}_0 = 0. \quad (9.33)$$

Applying the reflection symmetry to this geodesic produces another geodesic with exactly the same initial position and tangent vector. By uniqueness of the geodesic equation, the reflected geodesic must coincide with the original one. The trajectory therefore remains invariant under the reflection and hence satisfies

$$\boxed{\theta(\lambda) = \frac{\pi}{2}} \quad (9.34)$$

throughout its evolution.

The same result can be expressed using the separated geodesic equations. For the standard definition of the Carter constant $\mathcal{Q}$, equatorial initial conditions imply

$$\mathcal{Q} = 0, \quad (9.35)$$

and the polar potential satisfies

$$\Theta\left(\frac{\pi}{2}\right) = 0. \quad (9.36)$$

Thus $\mathcal{Q} = 0$ is not an additional independent condition for the equatorial initial data used here; it is the corresponding value of the Carter constant for a geodesic confined to the symmetry plane.

The restriction to equatorial initial conditions is primarily a controlled simplification of the Web Demo. The code sets

$$\theta_0 = \frac{\pi}{2}, \quad v_0^\theta = 0, \quad (9.37)$$

but it still evolves the complete eight-dimensional state

$$Y = (t, r, \theta, \phi, v^t, v^r, v^\theta, v^\phi) \quad (9.38)$$

using the full Kerr–Newman metric and the complete geodesic equation. Therefore θ is not manually reset to $\pi/2$ during the integration. Instead, the numerical quantity

$$\max_{\lambda} \left| \theta(\lambda) - \frac{\pi}{2} \right| \quad (9.39)$$

is monitored as a consistency check: in the exact theory it should vanish, while a small nonzero value measures numerical drift.

The advantage of this choice is that the dynamics can be visualized naturally through the two coordinates $r(\lambda)$ and $\phi(\lambda)$ while retaining the full tensorial geodesic calculation underneath. The price is that the current Web Demo does not explore generic Kerr–Newman geodesics with polar motion. Such trajectories generally have $\mathcal{Q} \neq 0$ and exhibit non-trivial evolution in θ . Consequently, the present simulator does not display inclined orbits, latitudinal oscillations, or the full three-dimensional structure of generic photon and massive-particle trajectories. These require independent non-equatorial initial data and a three-dimensional visualization and belong to the scope of the future Full Simulator.

9.5 Construction of the initial radial velocity

After computing $\dot{t}$ and $\dot{\phi}$ from (9.19)–(9.20), the solver sets $\dot{\theta} = 0$ and uses the normalization (9.13):

$$g_{rr}\dot{r}^2 = -\varepsilon - \left(g_{tt}\dot{t}^2 + 2g_{t\phi}\dot{t}\dot{\phi} + g_{\phi\phi}\dot{\phi}^2\right). \quad (9.40)$$

Therefore

$$\dot{r} = \pm \sqrt{\frac{-\varepsilon - N_{t\phi}}{g_{rr}}}, \quad N_{t\phi} = g_{tt}\dot{t}^2 + 2g_{t\phi}\dot{t}\dot{\phi} + g_{\phi\phi}\dot{\phi}^2. \quad (9.41)$$

The sign is chosen by the user as the ingoing or outgoing direction. If the radicand is negative, E , L_z , and r_0 do not define a real radial velocity: the simulator rejects the initial condition.

For intuition

Specifying r_0 , E , L_z , and the radial direction does not mean freely choosing all four components of the four-velocity. The two Killing constants fix $\dot{t}$ and $\dot{\phi}$; normalization fixes the magnitude of $\dot{r}$. Only the choice of sign remains.

Distinguished orbits and physical presets

10.1 Schwarzschild: photon sphere and ISCO

Before discussing rotation, it is useful to recall the structure of circular geodesics in Schwarzschild spacetime. In the limit

$$a = Q = 0, \quad (10.1)$$

the Kerr–Newman metric reduces to

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (10.2)$$

Because of spherical symmetry, any geodesic can be chosen to lie in a fixed plane. We choose the equatorial plane,

$$\theta = \frac{\pi}{2}, \quad \dot{\theta} = 0. \quad (10.3)$$

The metric then becomes

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\phi^2. \quad (10.4)$$

Stationarity and axial symmetry imply the conservation of

$$E = \left(1 - \frac{2M}{r}\right) \dot{t}, \quad L_z = r^2 \dot{\phi}. \quad (10.5)$$

The first quantity is the conserved energy per unit mass for a massive particle, or the corresponding conserved Killing energy for a photon; the second is the conserved axial angular momentum.

For a photon the trajectory is null, so

$$g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0. \quad (10.6)$$

Substituting the equatorial Schwarzschild metric gives

$$0 = -\left(1 - \frac{2M}{r}\right) \dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2. \quad (10.7)$$

Using the conserved quantities

$$\dot{t} = \frac{E}{1 - 2M/r}, \quad \dot{\phi} = \frac{L_z}{r^2}, \quad (10.8)$$

one obtains

$$\boxed{\dot{r}^2 = E^2 - \left(1 - \frac{2M}{r}\right) \frac{L_z^2}{r^2}}. \quad (10.9)$$

This equation describes the radial part of the photon trajectory. It is useful to rewrite it as

$$\dot{r}^2 + V_{\text{eff}}(r) = E^2, \quad (10.10)$$

where

$$\boxed{V_{\text{eff}}(r) = \left(1 - \frac{2M}{r}\right) \frac{L_z^2}{r^2}} \quad (10.11)$$

The analogy is with a one-dimensional mechanical problem: the conserved quantity E^2 plays the role of the total energy, while V_{eff} governs the allowed radial motion.

In particular,

$$\dot{r}^2 \geq 0 \quad (10.12)$$

requires

$$E^2 \geq V_{\text{eff}}(r). \quad (10.13)$$

Points at which

$$E^2 = V_{\text{eff}}(r) \quad (10.14)$$

are radial turning points, where $\dot{r} = 0$.

A circular photon orbit requires more than $\dot{r} = 0$. The photon must remain at the same radius, so the effective potential must also be stationary:

$$\dot{r} = 0, \quad \frac{dV_{\text{eff}}}{dr} = 0. \quad (10.15)$$

Since

$$V_{\text{eff}} = L_z^2 \left(\frac{1}{r^2} - \frac{2M}{r^3} \right), \quad (10.16)$$

one finds

$$\begin{aligned} \frac{dV_{\text{eff}}}{dr} &= L_z^2 \left(-\frac{2}{r^3} + \frac{6M}{r^4} \right) \\ &= \frac{2L_z^2}{r^4} (3M - r). \end{aligned} \quad (10.17)$$

Therefore the circular null orbit is located at

$$\boxed{r_\gamma = 3M}. \quad (10.18)$$

The condition $\dot{r} = 0$ at $r = 3M$ then gives

$$E^2 = \left(1 - \frac{2}{3}\right) \frac{L_z^2}{9M^2} = \frac{L_z^2}{27M^2}, \quad (10.19)$$

and hence

$$\boxed{\left| \frac{L_z}{E} \right| = 3\sqrt{3} M}. \quad (10.20)$$

The ratio

$$b = \frac{L_z}{E} \quad (10.21)$$

is the photon impact parameter. The two signs correspond simply to the two possible azimuthal orientations, which are physically equivalent in Schwarzschild because the spacetime is spherically symmetric and non-rotating.

The orbit at $r = 3M$ is unstable. Indeed,

$$\left. \frac{d^2 V_{\text{eff}}}{dr^2} \right|_{r=3M} < 0, \quad (10.22)$$

so the circular orbit lies at a maximum of the effective potential. An arbitrarily small radial perturbation therefore drives the photon away from $r = 3M$: depending on the perturbation and the conserved quantities, it is either captured by the black hole or escapes towards larger radii.

The sphere $r = 3M$ is therefore called the *photon sphere*. The circular orbit discussed above is its equatorial section; spherical symmetry makes all orbital planes physically equivalent.

For a massive particle, the normalization condition is instead

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = -1. \quad (10.23)$$

Using the same conserved quantities, the radial equation becomes

$$\dot{r}^2 = E^2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L_z^2}{r^2}\right). \quad (10.24)$$

Thus the massive-particle effective potential is

$$V_{\text{eff}}^{\text{massive}}(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{L_z^2}{r^2}\right). \quad (10.25)$$

For a circular orbit,

$$\dot{r} = 0, \quad \frac{dV_{\text{eff}}^{\text{massive}}}{dr} = 0. \quad (10.26)$$

Solving these conditions gives

$$\boxed{E = \frac{1 - 2M/r}{\sqrt{1 - 3M/r}}, \quad L_z = \pm \frac{\sqrt{Mr}}{\sqrt{1 - 3M/r}}.} \quad (10.27)$$

The sign of L_z only distinguishes the two possible directions of rotation; in Schwarzschild they are physically equivalent.

Circular timelike geodesics exist for

$$r > 3M, \quad (10.28)$$

but they are not all stable. Stability requires the circular orbit to correspond to a minimum of the effective potential,

$$\frac{d^2V_{\text{eff}}^{\text{massive}}}{dr^2} > 0. \quad (10.29)$$

This gives

$$r > 6M. \quad (10.30)$$

The limiting orbit

$$\boxed{r_{\text{ISCO}} = 6M} \quad (10.31)$$

is the *innermost stable circular orbit* (ISCO). Circular timelike geodesics with

$$3M < r < 6M \quad (10.32)$$

exist, but are unstable.

10.2 Kerr: prograde and retrograde orbits

When the black hole rotates, the spherical symmetry of Schwarzschild is lost. The Kerr spacetime is only stationary and axisymmetric, and the black hole possesses a preferred direction: the direction of its angular momentum

$$J = Ma. \quad (10.33)$$

As a consequence, the two possible orientations of an equatorial orbit are no longer physically equivalent.

An equatorial orbit is called *prograde* if its conserved axial angular momentum is aligned with the angular momentum of the black hole, and *retrograde* if it is anti-aligned. Equivalently,

$$\boxed{J L_z > 0 \quad \text{prograde}, \quad J L_z < 0 \quad \text{retrograde}.} \quad (10.34)$$

If one chooses $a > 0$, this reduces to

$$L_z > 0 \quad \text{prograde}, \quad L_z < 0 \quad \text{retrograde}. \quad (10.35)$$

This distinction is a direct consequence of frame dragging. The rotation of the black hole drags the geometry in the azimuthal direction. A prograde orbit moves in the same sense as this dragging and can remain circular at smaller radii. A retrograde orbit moves against it and requires a larger radius to remain on the corresponding circular orbit.

This should not be interpreted simply as the sign of the coordinate angular velocity $\dot{\phi}$. Near a rotating black hole, and especially inside the ergoregion, frame dragging affects the sign and magnitude of coordinate angular velocities. The invariant physical distinction is instead whether the orbital angular momentum is aligned or anti-aligned with the black-hole spin.

Introduce the dimensionless spin magnitude

$$\hat{a} = \frac{|a|}{M}, \quad 0 \leq \hat{a} \leq 1. \quad (10.36)$$

The use of $|a|$ is convenient because the formulas below distinguish the two orbital orientations explicitly as prograde and retrograde.

Define

$$\begin{aligned} Z_1 &= 1 + (1 - \hat{a}^2)^{1/3} \left[(1 + \hat{a})^{1/3} + (1 - \hat{a})^{1/3} \right], \\ Z_2 &= \sqrt{3\hat{a}^2 + Z_1^2}. \end{aligned} \quad (10.37)$$

The equatorial ISCO radii are

$$\boxed{\frac{r_{\text{ISCO}}^{\text{pro/retro}}}{M} = 3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}} \quad (10.38)$$

Here the upper choice of the sign,

$$-\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \quad (10.39)$$

gives the prograde ISCO, while the lower choice,

$$+\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \quad (10.40)$$

gives the retrograde ISCO.

Hence, for non-zero spin,

$$r_{\text{ISCO}}^{\text{pro}} < 6M < r_{\text{ISCO}}^{\text{retro}}. \quad (10.41)$$

Rotation moves the innermost stable prograde orbit inward and the innermost stable retrograde orbit outward.

The same asymmetry appears for equatorial circular photon orbits. Their radii are

$$\begin{aligned} \frac{r_{\gamma}^{\text{pro}}}{M} &= 2 \left[1 + \cos \left(\frac{2}{3} \arccos(-\hat{a}) \right) \right], \\ \frac{r_{\gamma}^{\text{retro}}}{M} &= 2 \left[1 + \cos \left(\frac{2}{3} \arccos(\hat{a}) \right) \right]. \end{aligned} \quad (10.42)$$

These are the rotating analogues of the Schwarzschild circular photon orbit at $r = 3M$. Unlike Schwarzschild, however, there is no single photon-sphere radius shared by every direction. Rotation breaks spherical symmetry and distinguishes the prograde and retrograde equatorial light rings.

For

$$\hat{a} = 0, \quad (10.43)$$

one has

$$Z_1 = 3, \quad Z_2 = 3, \quad (10.44)$$

and therefore

$$r_{\text{ISCO}}^{\text{pro}} = r_{\text{ISCO}}^{\text{retro}} = 6M. \quad (10.45)$$

Similarly,

$$r_{\gamma}^{\text{pro}} = r_{\gamma}^{\text{retro}} = 3M. \quad (10.46)$$

The distinction between the two orientations disappears, as expected in the Schwarzschild limit.

At extremal spin,

$$|a| = M, \quad (10.47)$$

the asymmetry becomes maximal. For the orientation aligned with the spin,

$$r_{\text{ISCO}}^{\text{pro}} \rightarrow M, \quad r_{\gamma}^{\text{pro}} \rightarrow M, \quad (10.48)$$

which coincides with the Boyer–Lindquist radius of the extremal horizon,

$$r_+ = M. \quad (10.49)$$

For the retrograde orientation,

$$r_{\gamma}^{\text{retro}} \rightarrow 4M, \quad r_{\text{ISCO}}^{\text{retro}} \rightarrow 9M. \quad (10.50)$$

Thus rotation creates a strong asymmetry between the two orbital directions: frame dragging allows prograde circular motion to extend much closer to the horizon, whereas retrograde circular orbits are displaced towards larger radii.

10.3 What the Web Demo does and does not show

The demo reports:

- Schwarzschild: $r_{\text{ISCO}} = 6M$, $r_{\gamma} = 3M$;
- Reissner–Nordström: the outer photon radius derived in the preceding chapter;

- Kerr: prograde/retrograde ISCOs and photon orbits from (10.38)–(10.42);
- general Kerr–Newman: “not available” for these orbits in the Web version.

The prograde/retrograde distinction in the code uses $|a|$; the names are relative to the spin direction. The sign of a determines which coordinate direction of ϕ is aligned.

10.4 Simulation presets

All presets use $M = 1$ in the solver and, unless otherwise stated, $Q = 0$.

Preset	object	a_*	r_0/M	E	L_z/M	direction
Circular photon	photon	0	3	1	$3\sqrt{3}$	outgoing ¹
Photon capture	photon	0.7	12	1	2.5	ingoing
Photon scattering	photon	0	18	1	6	ingoing
Circular massive orbit	massive	0	8	$\frac{1-2/8}{\sqrt{1-3/8}}$	$\frac{\sqrt{8}}{\sqrt{1-3/8}}$	outgoing
Massive infall	massive	0.7	12	1	0	ingoing
Massive flyby	massive	0	18	1.02	5	ingoing

The “circular massive orbit” preset uses (10.27) directly at $r = 8M$. Presets are not proofs that a stable orbit exists in every geometry: they are initial conditions chosen to illustrate specific behaviours.

¹The radicand of $\dot{r}$ vanishes: the direction does not alter the exact circular state.

From the tensor to the numerical code

11.1 From the continuous problem to the numerical state

The geodesic equations form a continuous differential system, whereas the browser can retain only a finite sequence of states. The translation implemented in the repository follows this chain:

1. convert M_{SI} , Q_{SI} , and J_{SI} into geometrized lengths and normalize $M = 1$;
2. compute Σ , Δ , and the components $g_{\mu\nu}$ at the current point;
3. numerically invert the metric and construct $\Gamma^\mu_{\alpha\beta}$;
4. derive the initial state from r_0 , E , L_z , object type, and radial direction;
5. rewrite the four second-order geodesic equations as eight first-order equations;
6. advance the state with RK4 and monitor normalization, energy, angular momentum, and confinement to the equatorial plane at every step.

This sequence separates the analytical results used to construct and check the state from the numerical approximation used to evolve it.

11.2 Construction of the metric

The physics core computes Σ , Δ , and $2Mr - Q^2$, then fills the symmetric matrix $g_{\mu\nu}$ according to (7.7). The inverse is not replaced by the analytical formula (7.9): it is obtained numerically by Gauss–Jordan elimination with pivoting. The analytical formula remains an important independent check.

Because the metric is stationary and axisymmetric,

$$\partial_t g_{\mu\nu} = 0, \quad \partial_\phi g_{\mu\nu} = 0. \quad (11.1)$$

The only nonzero derivatives are with respect to r and θ . The code computes them analytically from

$$\begin{aligned} \partial_r \Sigma &= 2r, & \partial_\theta \Sigma &= -2a^2 \sin \theta \cos \theta, \\ \partial_r \Delta &= 2(r - M), & \partial_\theta \Delta &= 0, \\ \partial_r (2Mr - Q^2) &= 2M. \end{aligned} \quad (11.2)$$

11.3 Christoffel symbols

The connection is not precomputed as a long list of formulas. At each point of the trajectory it is reconstructed from the definition

$$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta}). \quad (11.3)$$

Symmetry in the lower indices halves the independent pairs. The demo can list the numerically nonzero symbols at the initial point, grouping $\Gamma^\mu_{\alpha\beta} = \Gamma^\mu_{\beta\alpha}$. The display threshold is 10^{-9} ; this is not a symbolic claim that smaller quantities vanish exactly. Moreover, Christoffel symbols are not tensor components: their values depend on the chart, although the complete geodesic equation is covariant.

11.4 First-order system

The integrated state contains eight variables:

$$\mathbf{y} = (t, r, \theta, \phi, v^t, v^r, v^\theta, v^\phi), \quad v^\mu = \frac{dx^\mu}{d\lambda}. \quad (11.4)$$

The geodesic equation becomes

$$\begin{aligned} \frac{dx^\mu}{d\lambda} &= v^\mu, \\ \frac{dv^\mu}{d\lambda} &= -\Gamma^\mu_{\alpha\beta} v^\alpha v^\beta. \end{aligned} \quad (11.5)$$

This approach directly integrates the four second-order equations. It does not explicitly use the separated potentials (9.25)–(9.26); instead, E , L_z , and normalization are monitored as checks.

11.5 Fourth-order Runge–Kutta method

Write the first-order system as $\dot{\mathbf{y}} = F(\mathbf{y})$. With step size h , RK4 evaluates the direction field at the beginning, at two intermediate points, and at the end of the step:

$$\begin{aligned} k_1 &= F(\mathbf{y}_n), \\ k_2 &= F(\mathbf{y}_n + \tfrac{h}{2}k_1), \\ k_3 &= F(\mathbf{y}_n + \tfrac{h}{2}k_2), \\ k_4 &= F(\mathbf{y}_n + hk_3), \\ \mathbf{y}_{n+1} &= \mathbf{y}_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \quad (11.6)$$

The final combination assigns weight $1/6$ to the endpoint evaluations and $1/3$ to the intermediate ones. For a sufficiently regular solution, the local error is of order h^5 and the global error of order h^4 . The Web Demo uses a fixed step $h = 0.01$. It is simple and transparent, but neither symplectic nor adaptive: near regions of rapid variation, drift in the constants can grow, and formal order does not replace an actual error estimate.

11.6 Actual numerical parameters

Quantity	value	meaning
RK4 step	0.01	increment of the normalized affine parameter/proper time
default interval	80	maximum limit unless replaced by a preset
maximum steps	12000	protection against excessive cost
sampling	every 4 steps	points retained for the canvas
horizon margin	$0.12M$	stopping before r_+
singular radius	$0.05M$	stopping when no horizon exists

The maximum preset intervals range from 40 to 100. Since $M = 1$, all these quantities are dimensionless in the calculation.

11.7 Conservation checks

At every step, the following are reconstructed:

$$\begin{aligned}
 E &= -(g_{tt}v^t + g_{t\phi}v^\phi), \\
 L_z &= g_{t\phi}v^t + g_{\phi\phi}v^\phi, \\
 \mathcal{N} &= g_{\mu\nu}v^\mu v^\nu.
 \end{aligned} \tag{11.7}$$

The demo reports:

- the maximum absolute normalization error relative to the initial value;
- the maximum relative variation in E ;
- the maximum relative variation in L_z , or the absolute variation if $L_z = 0$;
- the maximum equatorial drift $|\theta - \pi/2|$.

These checks do not automatically make a trajectory correct: they test necessary conditions and allow integration quality to be diagnosed.

11.8 Stopping criteria

Integration terminates when it

1. reaches the maximum interval;
2. approaches the outer horizon to within $0.12M$;
3. reaches $r \leq 0.05M$ in a horizonless regime;
4. encounters a numerically non-invertible or singular metric;
5. encounters non-finite values or $r \leq 0$.

The demo does not continue through r_+ , analyse the region between the horizons, or simulate mass inflation at the inner horizon.

Physical guide to the simulator interface

The interface brings together quantities belonging to different levels: physical SI data, dimensionless geometrical parameters, constants of motion, and purely numerical quantities. To avoid ambiguity, the following sections follow the same order as the calculation: first the black hole is constructed, then the initial condition is assigned, the derived properties are read, and finally the canvas projection is interpreted.

12.1 Black-hole parameters

Entry	formula	interpretation
Mass M	$M_g = GM_{\text{SI}}/c^2$	global length and time scale
Charge Q	$Q_g = \sqrt{G/(4\pi\epsilon_0)}Q_{\text{SI}}/c^2$	source of the EM field; the metric depends on Q^2
Angular momentum J	$J_g = GJ_{\text{SI}}/c^3$	asymptotic rotational charge
Parameter a	$a = J_g/M_g$	spin length
Spin a_*	a/M_g	dimensionless spin; equal to a in the solver
Charge q_*	Q_g/M_g	dimensionless charge; equal to Q in the solver
Extremality	$M^2 - a^2 - Q^2$	positive: two horizons; zero: extremal; negative: no horizon

12.2 Initial conditions

Entry	solver units	meaning
r_0/M	dimensionless	initial Boyer–Lindquist coordinate radius
E	specific; dimensionless	constant associated with ∂_t
L_z/M	dimensionless	specific axial momentum associated with ∂_ϕ
$b = L_z/E$	multiples of M	impact parameter for null geodesics
direction	sign of $\dot{r}$	ingoing or outgoing
object	$\varepsilon = 0, 1$	null photon or neutral massive particle

12.3 Properties calculated in real time

The interface automatically classifies the metric according to the presence of Q and J . It displays:

- r_+ and r_- from (8.7);
- the *equatorial* static limit from (8.44);
- the discriminant $M^2 - a^2 - Q^2$;
- ISCOs and photon orbits in the implemented cases;
- the complete metric specialized to Schwarzschild, Reissner–Nordström, Kerr, or Kerr–Newman.

12.4 Visualization: reading the plot correctly

Each numerical point (r, ϕ) is drawn as

$$x = r \cos \phi, \quad y = r \sin \phi. \quad (12.1)$$

This transformation visualizes the equatorial coordinate trajectory. In general, it preserves neither physical distances, angles, nor areas. In particular:

- a circle on the canvas does not prove that the corresponding surface is a Euclidean sphere;
- the thickness of the ergoregion is a thickness in r , not a proper distance;
- the trajectory is not the optical image observed by a camera: that would require ray tracing with observer tetrads.

12.5 What is not yet simulated

The Web Demo does not include:

- charged test particles and the Lorentz force;
- generic three-dimensional geodesics with $Q \neq 0$;
- horizon crossing in regular coordinates;
- ray tracing, the shadow, gravitational lensing, and an accretion disk;
- backreaction of the particle or photon on the geometry;
- adaptive or symplectic integration and rigorous global-error estimates.

Test-particle regime

The mass, charge, and energy of the tracked object are assumed too small to modify the black hole's M , Q , or J . The simulator evolves a trajectory on a fixed background; it does not solve Einstein's equations coupled to the motion of the object.

Essential algebraic checks

13.1 From the compact squares to the components

Expand the two t - ϕ terms of (7.3):

$$\begin{aligned} -\frac{\Delta}{\Sigma}(\mathrm{d}t - a \sin^2 \theta \mathrm{d}\phi)^2 &= -\frac{\Delta}{\Sigma} \mathrm{d}t^2 + \frac{2a\Delta \sin^2 \theta}{\Sigma} \mathrm{d}t \mathrm{d}\phi - \frac{a^2 \Delta \sin^4 \theta}{\Sigma} \mathrm{d}\phi^2, \\ \frac{\sin^2 \theta}{\Sigma}[(r^2 + a^2) \mathrm{d}\phi - a \mathrm{d}t]^2 &= \frac{a^2 \sin^2 \theta}{\Sigma} \mathrm{d}t^2 - \frac{2a(r^2 + a^2) \sin^2 \theta}{\Sigma} \mathrm{d}t \mathrm{d}\phi \\ &\quad + \frac{(r^2 + a^2)^2 \sin^2 \theta}{\Sigma} \mathrm{d}\phi^2. \end{aligned} \quad (13.1)$$

The time coefficient is

$$-\frac{\Delta - a^2 \sin^2 \theta}{\Sigma} = -\frac{\Sigma - 2Mr + Q^2}{\Sigma} = -\left(1 - \frac{2Mr - Q^2}{\Sigma}\right). \quad (13.2)$$

The mixed coefficient is

$$-\frac{2a(r^2 + a^2 - \Delta) \sin^2 \theta}{\Sigma} = -\frac{2a(2Mr - Q^2) \sin^2 \theta}{\Sigma}. \quad (13.3)$$

This recovers the implemented form (7.5).

13.2 Determinant of the t - ϕ block

A useful identity is

$$g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -\Delta \sin^2 \theta. \quad (13.4)$$

Multiplying by $g_{rr}g_{\theta\theta} = \Sigma^2/\Delta$,

$$\det(g_{\mu\nu}) = (g_{tt}g_{\phi\phi} - g_{t\phi}^2)g_{rr}g_{\theta\theta} = -\Sigma^2 \sin^2 \theta, \quad (13.5)$$

in agreement with (7.10).

13.3 Why the horizon does not coincide with $g_{tt} = 0$

In a static spherical metric, $g_{tt} = 0$ and $g^{rr} = 0$ occur on the same surface. In Kerr–Newman,

$$g^{rr} = 0 \iff \Delta = 0, \quad g_{tt} = 0 \iff \Delta - a^2 \sin^2 \theta = 0. \quad (13.6)$$

The two conditions differ by $a^2 \sin^2 \theta$. They coincide at the poles or when $a = 0$, but not at the equator of a rotating black hole. Confusing the two surfaces means confusing the event horizon with the ergosurface.

Quick reference for units

Quantity	from SI to geometrized	dimensionless with $M = 1$
Mass	$M_g = GM_{\text{SI}}/c^2$	$M/M_g = 1$
Time	$t_g = ct_{\text{SI}}$	t/M_g
Radius	$r_g = r_{\text{SI}}$	r/M_g
Charge	$Q_g = \sqrt{G/(4\pi\epsilon_0)}Q_{\text{SI}}/c^2$	$q_* = Q_g/M_g$
Angular momentum	$J_g = GJ_{\text{SI}}/c^3$	J_g/M_g^2
Kerr parameter	$a = J_{\text{SI}}/(M_{\text{SI}}c)$	$a_* = a/M_g$
Angular velocity	$\Omega_{\text{SI}} = c\Omega_g$	$M_g\Omega_g$

Map between theory and implementation

Theoretical object	Formula	Use in the code
Radial function	$\Delta = r^2 - 2Mr + a^2 + Q^2$	metric, horizons, extremality
Angular function	$\Sigma = r^2 + a^2 \cos^2 \theta$	all metric components
Compact source	$2Mr - Q^2$	$g_{tt}, g_{t\phi}, g_{\phi\phi}$
Horizons	$r_{\pm} = M \pm \sqrt{M^2 - a^2 - Q^2}$	properties and canvas circles
Equatorial static limit	$M + \sqrt{M^2 - Q^2}$	outer edge of the ergoregion on the canvas
Constants	$E = -p_t, L_z = p_\phi$	initial state and checks
Normalization	$g(v, v) = -1, 0$	calculation of v_0^r and drift
Connection	equation (11.3)	geodesic acceleration
Geodesic	equation (11.5)	eight-component system
Integrator	RK4, equation (11.6)	fixed-step numerical evolution
Projection	$x = r \cos \phi, y = r \sin \phi$	equatorial visualization

The implementation analysis refers to the website repository, `main` branch, commit 0036b85a, verified on 18 August 2026 [12].

Minimal glossary

Asymptotically flat

Spacetime approaches the Minkowski metric at large distance.

Electrovacuum

A region without material current J^μ , but with an electromagnetic field and its associated $T_{\mu\nu}$.

Ergoregion

A region where ∂_t is spacelike and no observer can remain static relative to infinity.

Extremal

A configuration $M^2 = a^2 + Q^2$ with a double horizon and $\kappa = 0$.

Null geodesic

A trajectory with $g(v, v) = 0$, appropriate to the geometrical-optics propagation of photons.

Timelike geodesic

A trajectory with $g(u, u) = -1$, appropriate to a freely falling neutral massive particle.

Cauchy horizon

A boundary beyond which exterior initial data do not uniquely determine the maximal evolution.

Event horizon

The causal boundary of the past of future null infinity; signals from inside cannot reach exterior infinity.

Areal radius

A coordinate r for which the symmetry spheres have area $4\pi r^2$; valid in Reissner–Nordström, but not generally in Kerr.

Stationary/non-static

Independent of t , but with rotational dragging $g_{t\phi} \neq 0$.

Conclusion

The entire construction can be summarized as a chain of consequences:

1. **Global charges.** M , Q , and J are not local labels: they are conserved quantities read at infinity or through fluxes across closed surfaces.
2. **Electromagnetic field.** Antisymmetry organizes the electric and magnetic fields into the 2-form F ; the homogeneous equation locally implies $F = dA$.
3. **Gravitational source.** Variation of the action with respect to the metric produces $T_{\mu\nu}^{\text{EM}}$, quadratic in F , traceless, and with positive energy density.
4. **Reissner–Nordström.** Static spherical symmetry reduces Einstein–Maxwell to integrable radial equations and generates

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}.$$

5. **Two horizons, two roles.** r_+ is the outer event horizon; r_- is an inner Cauchy horizon, unstable to blueshift and mass inflation.
6. **Rotation.** Angular momentum breaks spherical symmetry, leaves stationarity and axisymmetry, and introduces the physical term $g_{t\phi}$.
7. **Boyer–Lindquist.** The transformation from advanced coordinates removes the mixed radial terms and makes symmetries and constants of motion manifest, but contains $1/\Delta$ and fails on the horizons.
8. **Kerr–Newman.** The functions

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad \Delta = r^2 - 2Mr + a^2 + Q^2$$

respectively separate angular and radial structure; $\Delta = 0$ determines the horizons, whereas $g_{tt} = 0$ determines the static limit.

9. **Geodesics.** $E = -p_t$, $L_z = p_\phi$, normalization, and equatorial symmetry construct the initial state of the neutral particle.
10. **Simulation.** The code normalizes $M = 1$, constructs the metric in Boyer–Lindquist coordinates, integrates the eight-component system with RK4, and uses conserved quantities as numerical diagnostics.

Central message

The simulator does not display an object embedded in Euclidean space: it displays coordinates of causal trajectories in a geometry determined by M , Q , and J . Horizons, the ergoregion, numerical stopping conditions, and canvas circles acquire meaning only if the physical geometry is kept distinct from the Boyer–Lindquist chart and the graphical projection.

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